

Nephrolithiasis

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Nephrolithiasis (aliases: kidney stone disease, urolithiasis) is the abnormal formation and retention of solid-phase inorganic and organic concretions in the lumen of the urinary tract. Nephrolithiasis is not a diagnosis per se but the common manifestation of a variety of underlying causative and pathophysiologic factors. Although stones are localized to the urinary tract, nephrolithiasis is a systemic disorder. The surgical treatment of nephrolithiasis has greatly advanced in recent years, yet the management extends beyond the removal of existing stones. A careful metabolic evaluation in kidney stone formers reveals the underlying pathophysiology and systemic disease manifestations and forms the basis for individualized recurrence prevention and the treatment of comorbid conditions.

EPIDEMIOLOGY

GENERAL POINTS

The prevalence of kidney stones has increased steadily in recent decades¹ and, consequently, so have both direct and indirect costs for this condition.² The prevalence of kidney stones is highest in non-Latin Caucasians, followed by Latin Caucasians and Asians, with the lowest incidence in African Americans.^{1,3} Age-adjusted prevalence is higher in southern parts of the United States.⁴ Potential factors contributing to regional differences include the quality of water supply, climate, animal protein intake, and comorbid conditions.

One estimate found that 1 in 11 individuals in the United States has a history of kidney stones.⁴ The overall prevalence was 10.6% in men and 7.1% in women (Fig. 40.1).³ This observation is not limited to the United States; a continuous increase in prevalence throughout the decades, with similar age and sex distribution profiles across countries, has been observed.⁵ In general, the prevalence of kidney stone disease in adults is higher in the Western Hemisphere.⁶ Of interest, the highest kidney stone prevalence has been observed in Saudi Arabia.⁶ The prevalence of kidney

stones in children, while significantly lower, also continues to rise.^{7,8} Interestingly, the group with the greatest rise was adolescent girls.⁸ Pediatric nephrolithiasis shows a different sex predilection; kidney stone incidence is higher in adolescent girls relative to boys, whereas there are no notable sex differences in prepubertal children.⁷

STONE COMPOSITION

The majority of stones in children and adults are composed of calcium, predominantly as calcium oxalate but to a lesser extent as calcium phosphate (brushite).^{9,10} In adults, the next most frequent stone type is uric acid (UA); this contrasts with children where up to 8% to 10% are composed of cystine.¹¹ There are regional differences in UA stone prevalence. The highest prevalence of UA stones is reported in the Middle East and a few European countries,¹²⁻¹⁴ whereas UA stones comprise only 8% to 10% of all kidney stones in the United States.¹⁵ Less common stone types in adults are cystine (1%–2%) and in children UA (2%–3%).¹⁶ The reason for this discrepancy is likely genetics, with monogenetic disorders such as cystinuria manifesting early in life, and lifestyle and dietary factors, as well as complex genetics affecting stone composition later in life.

DATA ON STONE INCIDENCE

Compared with prevalence, precise knowledge of kidney stone incidence among adult and pediatric populations is limited. In adult populations, one small study from Minnesota has shown an incidence rate of 101.8/100,000 patients per year.¹⁷ Two studies using a questionnaire found an overall incidence rate among adult males and females of 306 and 95/100,000 person-years, respectively.^{18,19}

The most recent reported incidence of pediatric nephrolithiasis in the United States is 65/100,000 person-years, which is a significant increase over the past 3 decades.^{7,8,20-22} It has been suggested that childhood obesity may play a key role in the rising incidence of kidney stones in

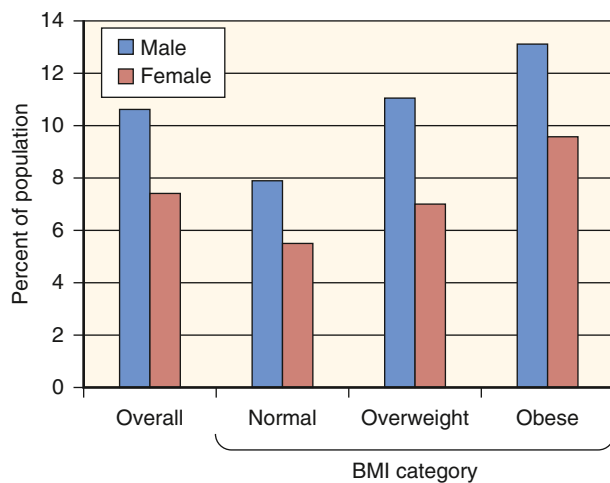


Fig. 40.1 Prevalence of urolithiasis plotted as a function of gender and body mass index (BMI). (Modified from Scales CD Jr, Smith AC, Hanley JM, Saigal CS. Urologic diseases in America. Prevalence of kidney stones in the United States. *Eur Urol.* 2012;62[1]:160–165.)

children. However, larger U.S. epidemiologic studies do not support this notion.²³ Dietary factors including increased sodium consumption²⁴ and decreased calcium intake²⁵ from milk, which has been substituted by sugar drinks,^{26,27} and reduced water intake²⁸ have also been suggested to have a pathophysiologic role in the development of kidney stones in children.

HISTOPATHOLOGY

The histopathology of nephrolithiasis that genuinely resembles human disease, involving both urinary and epithelial factors, is rather different than in rodents.²⁹ Thus the main source of human histopathologic data has been obtained from clinical surgical samples. In addition to urinary factors such as solutes, which comprise the chemical components of stones, known inhibitors and promoters, as well as epithelial factors that initiate and promote crystal adhesion, growth, and agglomeration, contribute.³⁰ The field has advanced significantly by the work of Evan and colleagues^{31,32} who used human samples from intraoperative biopsies to provide histologic descriptions, enabling the formulation of pathogenic models of stone formation.

RANDALL PLAQUE

In the 1930s, Randall examined the papilla of >1000 pairs of cadaveric kidneys and observed a cream-colored area near the papillae in approximately one in five kidneys. These plaques are mainly interstitial rather than luminal and associate with interstitial collagen and tubular basement membranes; they are composed of calcium, nitrogen, carbon dioxide, and phosphorus.³³ More advanced lesions intrude into tubular lumens. What differentiates these plaques from more normal variants was the finding of small stones attached to some of the plaques, projecting into the renal pelvis in more than 60 kidneys. Randall concluded that the attached stones were growing from the interstitial calcium plaque, rather than directly from the normal epithelium.³³ There was no clinical

information on Randall's subjects, but there is a reasonable chance that they were calcium oxalate stone formers. These observations have been confirmed by many surgeons, with the Indianapolis-Chicago group shedding light on the importance of Randall's plaques in the pathogenesis of stone formation. The pathogenesis of Randall's plaques remains an enigma, but a degenerative stress model of induction of dedifferentiation and transdifferentiation, followed by pathologic biomineralization, has been proposed.³⁴

IDIOPATHIC CALCIUM OXALATE STONES

Plaques form in the papillary tip in the basement membrane of the thin loop of Henle, with mineral and organic layers alternating in a concentric configuration (Fig. 40.2). Evan and coworkers³⁵ have proposed that this is the final form of plaque composed of fused particles in close association with type 1 collagen, with the mineral phase covered by organic matrix. In the absence of longitudinal biopsies, extrapolation from cross-sectional data suggests that the particles move from the tubular basement membrane into the surrounding interstitium.

En bloc examination of stones with attached tissue³⁶ shows loss of epithelial cells at the attachment site with the plaque being exposed to the urinary lumen. On contact with urine, the plaque is overlaid with new matrix, and new crystals and matrix form in sequential waves. Osteoprotegerin is present in both the stone and plaque, but Tamm-Horsfall protein, which is known to be restricted to the urine and cells of the thick ascending limb, is present only on the urine side of the interface.³⁶ It has been proposed that the organic material overlaying the exposed plaque comes from urine molecules adsorbed initially onto the matrix of plaque. As new crystals nucleate in this urine matrix, the crystals themselves attract molecules, thereby perpetuating stone formation.³⁷

CALCIUM PHOSPHATE STONES AND STONES IN RENAL TUBULAR ACIDOSIS

Although Randall plaques are observed in calcium phosphate (CaP) stone formers, they are fewer in number compared with calcium oxalate stone formers. CaP stone formers have apatite deposits in the lumen ducts of Bellini and inner medullary collecting ducts that are associated with massive ductal dilation (Fig. 40.3).³⁸ The dilated ducts appear to have lost epithelial cells and are surrounded by fibrotic tissue. The abundance of CaP crystals is correlated with higher urinary CaP supersaturation,³⁹ which is driven largely by a high urine pH and, to a lesser extent, by hypocitraturia and hypercalciuria. A scenario of persistently high urinary pH is seen in CaP stone formers with distal renal tubular acidosis (dRTA) secondary to genetic or acquired causes. Papillae show multiple dilated ducts of Bellini, with intraluminal CaP deposits. Distortion, flattening, and fibrosis are common with atrophic remnants of nephron structures lying within fibrotic fields of interstitium.

STONES IN ENTERIC HYPEROXALURIA

In patients with enteric hyperoxaluria, crystallization is mainly driven by a high urinary oxalate concentration. In patients post gastric bypass surgery, the renal tubular

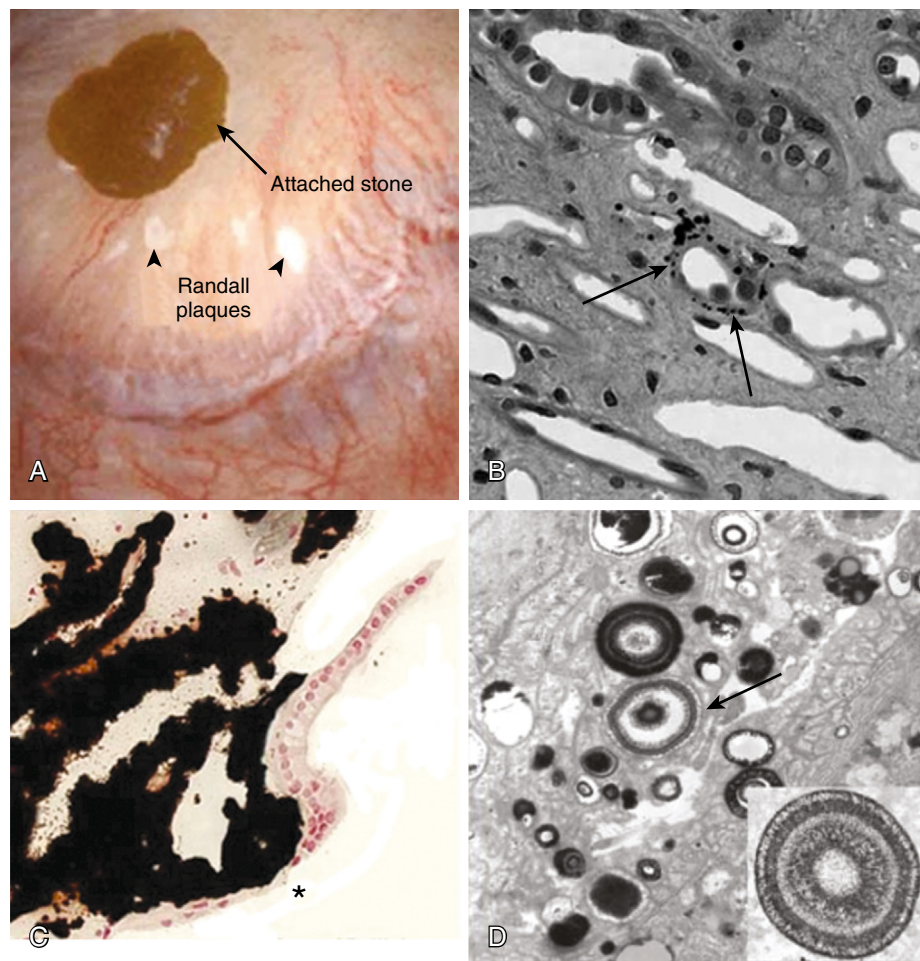


Fig. 40.2 Histopathology of Randall's plaques in calcium oxalate stone formers. (A) View during endoscopic surgery. One plaque (not visible) has an overlying attached calcium oxalate stone (arrow). Two Randall plaques without stones are below (arrowheads). (B) Light microscopic image of the locale of the initial crystal deposits in the basement membrane of the thin loops of Henle (arrows). (C) Calcium phosphate (stained black) showing a more advanced lesion filling the interstitial space and extending (*), eventually protruding into the urinary space and corresponding to the cream-colored plaques in A. (D) The individual deposits (arrow) assume the morphology of multilaminated spheres of alternating layers of electron-lucent inorganic crystal and electron-dense matrix (inset). (From Evan AP, Lingeman JE, Coe FL, et al. Randall's plaque of patients with nephrolithiasis begins in basement membranes of thin loops of Henle. *J Clin Invest.* 2003;111[5]:607–616.)

epithelium appears normal, but calcium oxalate crystals are lodged in the lumen.⁴⁰ In hyperoxaluric calcium oxalate stone formers due to small bowel resection (e.g., as a result of Crohn's disease), the inner medullary collecting ducts contain crystal deposits associated with cell injury, interstitial inflammation, and structural deformity of the papillae, tubular atrophy, and interstitial fibrosis. Interestingly, Randall plaques are observed, similar to those seen in idiopathic calcium oxalate stone formers. The inner medullary collecting duct deposits contain apatite, with calcium oxalate in some cases.⁴¹ The birefringent thin crystalline material scattered on the inner medullary collecting duct (IMCD) cell membranes may be the initial crystal lesion.

Overall, based mainly on histopathologic findings and aided by clinical chemistry and imaging, three morphologic pathways of stone formation have been proposed.³⁷ First is overgrowth on interstitial apatite plaque, as seen commonly but not exclusively in idiopathic calcium oxalate stone formers; similar lesions are observed in stone formers with primary hyperparathyroidism, ileostomy, and small

bowel resection and in some brushite stone formers. Second are crystal deposits in renal tubules rather than the interstitium, which are seen in all stone formers other than the idiopathic calcium oxalate stone formers. The third is free solution crystallization in the lumen, as seen in patients with cystinuria or enteric hyperoxaluria.

PATHOPHYSIOLOGY

PHYSICAL CHEMISTRY OF URINARY SATURATION

GENERAL CONCEPTS

Lithogenesis is a complex process that involves solutes entering a solid phase in urine. Nucleation is the first step and occurs when the concentration of a salt (e.g., calcium oxalate) in solution reaches a level at which a solid phase (e.g., calcium oxalate crystals) begins to appear. An aqueous solution containing calcium oxalate and calcium oxalate crystals is in equilibrium when crystals neither grow nor shrink. In this situation, the product of the free

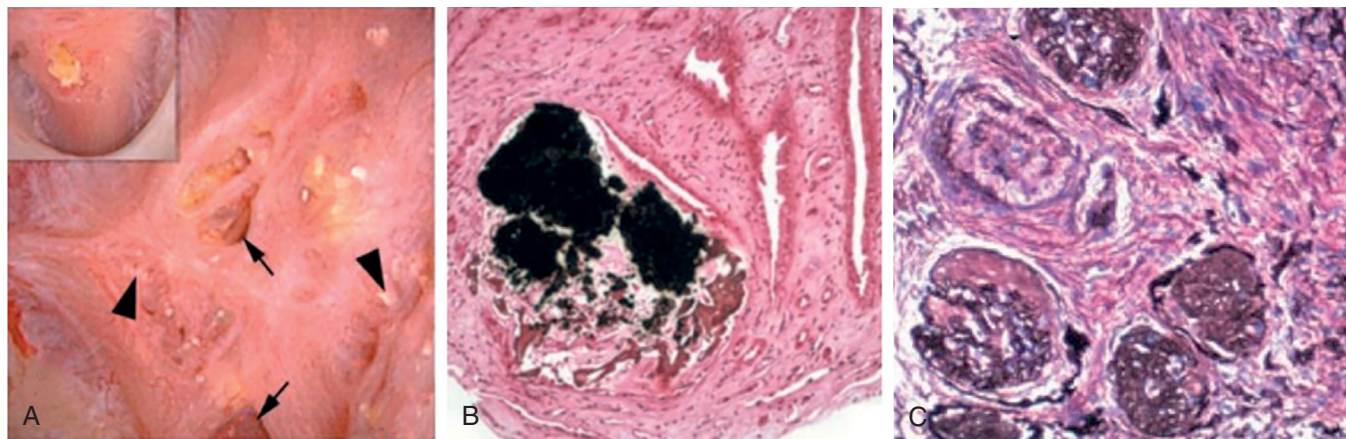


Fig. 40.3 Histopathology of calcium phosphate stone formers. (A) View during endoscopic surgery. Depressions near the papillary tips (arrows) are unique to calcium phosphate stone formers, which coexist with Randall plaques (arrow heads). The papillae show yellow crystalline deposits coming out of the ducts of Bellini (inset). (B) Micrograph of luminal deposits in an inner medullary collecting duct. The crystal deposits greatly expanded the lumen, and cell injury and necrosis were found. Interstitial inflammation and fibrosis surround the intraluminal crystal deposition. (C) Renal biopsy from a patient with calcium phosphate stones showing advanced glomerulosclerosis, tubular atrophy, and interstitial fibrosis. This is rarely seen in calcium oxalate stone formers. (From Evan AP, Lingeman JE, Coe FL, et al. Crystal-associated nephropathy in patients with brushite nephrolithiasis. *Kidney Int.* 2005;67[2]:576–591.)

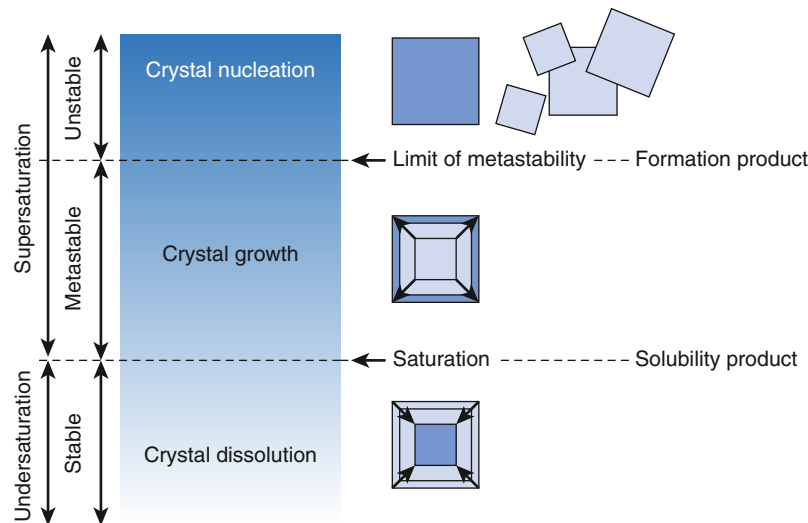


Fig. 40.4 Physicochemical parameters used in the assessment of kidney stone risk. These are shown in relation to the three states of crystal—dissolution, growth, and nucleation.

ionized calcium and oxalate concentrations in solution (activity product; AP) is equivalent to the equilibrium solubility product (SP). Activity refers to the portion of the ion in solution that is chemically active (activity = activity coefficient $\times$ chemical concentration). If $AP < SP$, crystals will dissolve; the solution is undersaturated. If $AP > SP$, the solution is supersaturated and crystals will grow (Fig. 40.4). An AP (e.g., calcium oxalate ion product) divided by the corresponding equilibrium SP yields the activity product ratio (APR), which estimates the degree of saturation. An $APR > 1$ indicates urinary supersaturation, and an $APR < 1$ indicates urinary undersaturation.

When calcium and oxalate are added to a saturated solution at equilibrium to raise the ion activity product beyond the equilibrium SP, there will be growth of any preformed crystals, if such were present. However, in the absence of a preexisting solid phase, no new crystals appear,

despite an AP above the SP. A solution that will cause the growth of preformed crystals but not the appearance of a new solid phase is supersaturated and metastable. If one raises the AP even higher, with further calcium and oxalate addition, new crystals will appear at some point without existing crystals (see Fig. 40.4). The AP has now reached the formation product (FP), which is also called the upper limit of metastability (ULM). Above the FP, a solution changes from metastable to unstable and crystallization is inevitable. Therefore urine may be undersaturated, metastable, or unstable with respect to lithogenic salts (see Fig. 40.4).

FACTORS INFLUENCING SATURATION

Rates of urinary excretion of the component species of the AP (e.g., calcium, oxalate, phosphate, and urate) and water (denominator) are the primary determinants of saturation. Additional factors, such as complexation and changes in

urine pH, can all influence the free ion concentrations and are important in regulating saturation. Therefore total concentration measurements may not provide complete information on the actual AP. For example, citrate readily complexes calcium, reducing the ionized calcium (activity) levels⁴²; a similar relationship exists for magnesium and oxalate.⁴³ Changes in urine pH can drastically affect the monovalent-to-divalent phosphate and urate-to-UA ratio. For this reason, hypercalciuria, hyperoxaluria, hypocitraturia, unduly alkaline urine, and chronic dehydration all increase the risk of calcium stones, but the relationships are complex.

Thus interpretation based solely on individual chemical concentrations can be misleading. An example is when a patient with idiopathic hypercalciuria is treated with thiazide and the urinary calcium excretion decreases. However, a state of potassium deficiency can develop, causing secondary hypocitraturia. It is not easy for a clinician to look at the urinary calcium and citrate excretion rates and concentrations and then determine whether the stone risk is reduced. Another scenario is a patient with calcium phosphate stones from distal renal tubular acidosis and profound hypocitraturia that is treated with alkali. The urinary citrate level increases, but the urine pH concomitantly increases as well, so it is not easy to determine whether the risk of calcium phosphate crystallization is reduced or increased. Several methods have been designed to address this.

URINE SATURATION ESTIMATION IN SILICO

ULM, FP, and the APR can be determined empirically. While these empiric physicochemical methods are informative and important tools in research, their obvious drawback is the labor involved and skills needed to perform these delicate measurements. Another option is to take clinical urine chemistry data and employ software to calculate urine-free ion activities for calcium, oxalate, and phosphate from their measured chemical concentrations and their known tendencies to form soluble complexes with each other and with other ligands.⁴⁴⁻⁴⁶

EQUIL2 Software: Relative Supersaturation Ratio

Using all the relevant association constants (K_a), one can calculate the AP from clinical routine urine chemistry data. If one divides this by the corresponding equilibrium SP, one obtains the relative supersaturation ratio (RSR), which estimates the degree of saturation. An $RSR > 1$ connotes oversaturation and < 1 , undersaturation. The validity has been confirmed by studies showing a correlation between the type of stone a patient forms and the prevailing supersaturations in 24-hour urine samples, as estimated by the EQUIL2 program. Further, interventions that reduced stone events in randomized controlled trials and prospective studies closely correlated with reductions in calculated urine RSRs.⁴⁷⁻⁵⁰

JESS Software: Supersaturation Index

Another software, the Joint Expert Speciation System (JESS), uses the same principle as EQUIL2 but differs from it in that more thermodynamic constants are used to calculate mixed ligand speciation.^{51,52} One important difference is that JESS includes the calcium phosphocitrate complex, which is soluble, and whose formation is pH dependent. Pak and

coworkers⁵³⁻⁵⁵ have compared the two programs to empiric physicochemical methods and found a closer approximation of JESS than EQUIL2 to physicochemical methods. While EQUIL2 provides RSR as a readout, JESS expresses the data as the solubility index (SI), which is calculated based on the same physicochemical principles as RSR values generated by EQUIL2, but the two programs use different speciation concentrations.

URINE SATURATION IN STONE FORMERS

Urine from stone formers is more supersaturated with respect to calcium oxalate, brushite, octocalcium phosphate (a unique form of calcium phosphate), or hydroxyapatite compared with urine from non-stone formers.⁵⁶⁻⁵⁹ Yet most urine from non-stone formers is supersaturated with respect to calcium oxalate.^{56,59} Hautmann and coworkers⁶⁰ measured calcium and oxalate concentrations in tissues from the cortex, medulla, and papilla of seven human kidneys. The calcium oxalate concentration product in the papillae ($1 \times 10^{-4} \text{ M}^2$) exceeded that of urine ($5 \times 10^{-7} \text{ M}^2$) and those of the medulla and cortex ($8 \times 10^{-7} \text{ M}^2$ and $6 \times 10^{-7} \text{ M}^2$, respectively). In addition, high calcium phosphate supersaturation is common in the tip of loop of Henle because both tubule fluid pH and increased calcium concentrations result from water extraction as tubular fluid descends the descending limb.⁶¹

The actual ULM for calcium oxalate and brushite in urine samples from non-stone formers and stone formers are surprisingly variable.⁵⁶ The ULM of calcium oxalate and brushite was measured in urine from calcium stone formers and gender- and age-matched control individuals^{62,63}; the gap between prevailing supersaturation of the urine and the ULM was reduced in stone formers, rendering it more likely for crystallization and stone formation to occur. The reduced ULM may represent deranged crystallization inhibition.

NUCLEATION, GROWTH, AND AGGREGATION

Saturation estimates the propensity to crystallize. Nucleation refers to the initial formation of a crystal nidus. Nucleation is followed by crystal growth, epitaxial growth, and aggregation. Homogeneous nucleation, the spontaneous formation of new crystal nuclei in a supersaturated metastable solution, is actually uncommon. Usually, debris particles, surface irregularities, or other existing crystals furnish a substrate on which crystal nuclei begin to form at a lower APR than what is required for true homogeneous nucleation. The metastable zone reflects the greater free energy required to create new nuclei than to simply enlarge preformed nuclei. Any surface that can serve as a substrate for ions in solution may act as a heterogeneous nucleus, bypassing the energetically more costly process of creating a solid phase de novo. In other words, there is a lower apparent ULM for heterogeneous nucleation.

The efficiency of heterogeneous nucleation depends on the similarity between the spacing of charged sites on the preformed surface and in the lattice of the crystal that is to grow on that surface. This matching is referred to as “epitaxis,” and its extent is usually referred to as a good or poor epitaxial relationship.⁶⁴ A number of urine crystals have good epitaxial matching and behave toward one another as heterogeneous nuclei. Monosodium urate and UA are

excellent heterogeneous nuclei for calcium oxalate.^{65,66} Therefore UA or urate could, by crystallizing, lower the ULM for calcium oxalate. Heterogeneous nucleation may be the mechanism linking hyperuricosuria to calcium oxalate stones.⁶⁷⁻⁶⁹ Both brushite and hydroxyapatite can also nucleate calcium oxalate,^{70,71} which is likely how Randall plaque acts as an initiator of calcium oxalate stones.

Growth is the enlargement of a crystal, and aggregation is the coalescence of crystals. Once formed, crystals will grow if bathed in urine with an APR >1. Growth and aggregation are pathogenic because microscopic nuclei are too small to cause disease. Crystals are regular lattices, composed of repeating subunits, and they grow by the incorporation of calcium and oxalate or phosphate, or UA, into new subunits on their surfaces. The growth rate increases with supersaturation and is most rapid in urine samples with the highest APR. Small crystals aggregate into larger crystalline masses by electrostatic attraction from the charged surface of the crystals, which rapidly increase particle size, producing a crystal that can lodge in the urinary tract. Stone formers' urine contains larger crystal aggregates compared with non-stone formers.⁷²

CALCIUM STONES

CaOx is the most prevalent kidney stone worldwide, accounting for approximately 70% to 80% of kidney stones. CaP stones contribute a smaller percentage (15%).⁷³⁻⁷⁷ Risk factors for CaOx stone formation include hypercalciuria, hyperuricosuria, hypocitraturia, and hyperoxaluria.^{73,74} CaP stones share some common risk factors with CaOx stones including hypercalciuria and hypocitraturia but, in contrast to CaOx, unduly alkaline urine is characteristic of CaP stone formers but not in CaOx stone formers. Next, we discuss these and other risk factors such as hyperuricosuria.

HYPERCALCIURIA

Hypercalciuria is the most prevalent metabolic abnormality in patients with calcium nephrolithiasis, occurring in up to

60% of adults with calcium stones and 45% of children.⁷⁶⁻⁸² The pathophysiologic mechanisms for hypercalciuria are multifold and may involve increased intestinal absorption (absorptive hypercalciuria; Fig. 40.5), diminished renal tubular calcium reabsorption (renal leak hypercalciuria), and enhanced calcium mobilization from the bone (resorptive hypercalciuria).^{74,83-86} Intestinal hyperabsorption of calcium is the most common abnormality observed in adult kidney stone formers.⁸⁷ Importantly, all or more than one of these physiologic derangements may coexist in individual patients.

INTESTINAL HYPERABSORPTION OF CALCIUM

Flocks first showed a link between hypercalciuria and nephrolithiasis.⁸⁸ Subsequently, Albright and Henneman^{89,90} have used the term "idiopathic hypercalciuria" to highlight the unknown origin of hypercalciuria in this population. Current clinical practice does not require one to determine the origin of hypercalciuria, given that therapies are the same. However, with the movement toward precision health and the potential for targeted therapies, it is worth reviewing the possible etiologies of this common risk factor for kidney stone formation.

The characteristic features of absorptive hypercalciuria are hypercalciuria with normocalcemia, normal or suppressed serum parathyroid hormone (PTH), and/or normal or suppressed urinary cyclic adenosine 3',5'-monophosphate (cAMP; a surrogate marker of PTH bioactivity in the kidney). These findings are consistent with an increased calcium load. Hyperabsorption is the most common feature of patients with idiopathic hypercalciuria, which can be divided into two broad subtypes, calcitriol-dependent⁹¹⁻⁹⁵ and calcitriol-independent.^{87,96-99}

Elevated 1,25(OH)₂D has been observed in multiple studies examining hypercalciuric nephrolithiasis patients.⁹¹⁻⁹⁴ The presence of absorptive hypercalciuria was established by direct measurement of intestinal calcium absorption^{98,100} or inferred⁹³ due to a lower serum PTH concentration. Both increased production and decreased clearance have been proposed as mechanisms responsible

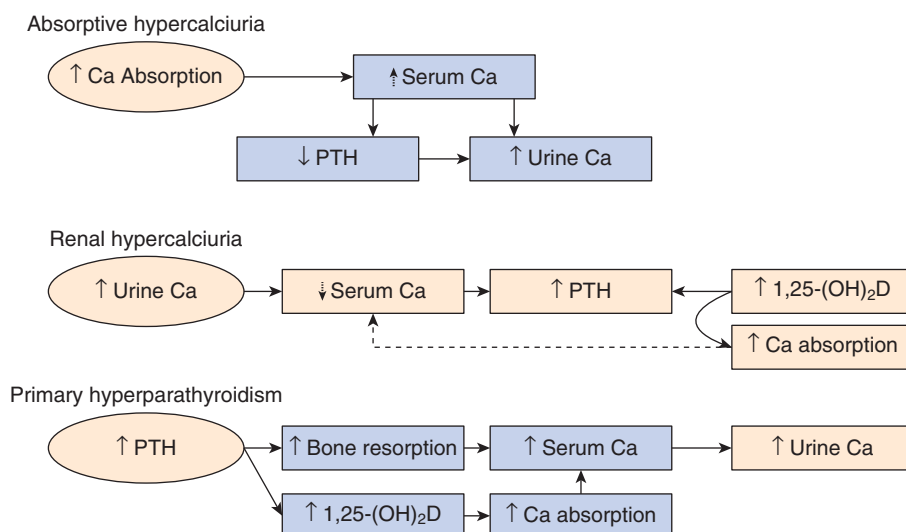


Fig. 40.5 Pathophysiologic mechanisms of hypercalciuria. PTH, Parathyroid hormone. (From Pak CY. Etiology and treatment of urolithiasis. *Am J Kidney Dis.* 1991;18[6]:624–637.)

for increased serum $1,25(\text{OH})_2\text{D}$, but the underlying molecular mechanisms remain elusive.^{92,101} Further evidence supporting $1,25(\text{OH})_2\text{D}$ enhancing urinary calcium excretion is provided by a study of normal subjects who became hypercalciuric after receiving a high dose of $1,25(\text{OH})_2\text{D}$.⁹¹ This is consistent with net intestinal calcium absorption significantly increasing after calcitriol administration and urinary calcium excretion of stone formers exceeding that of controls.¹⁰² There is also a direct correlation between serum $1,25(\text{OH})_2\text{D}$, calcium excretion, and calciuric responses, consistent with a pathogenetic role of excess $1,25(\text{OH})_2\text{D}$ in increasing both intestinal absorption and renal excretion.⁹¹ However, it is still unknown whether the calcium in $1,25(\text{OH})_2\text{D}$ -driven hypercalciuria originates from the bone¹⁰² or intestine.⁹¹

It should be noted that a small proportion of patients with absorptive hypercalciuria display a defect in renal tubular reabsorption of phosphorus (renal phosphorus wasting), which in turn stimulates $1,25(\text{OH})_2\text{D}$ synthesis and therefore intestinal calcium absorption.^{93,103,104} Emerging evidence from large kidney stone cohorts who have undergone unbiased genetic testing suggests that this may be due to monogenic variants in *SLC34A3*.¹⁰⁵ Biallelic pathogenic variants in *SLC34A3* (which encodes the sodium-dependent phosphate transporter 2C) cause a rare condition characterized by hypophosphatemic rickets with hypercalciuria.¹⁰⁶ Carriers of monoallelic variants in *SLC34A3* display a milder phenotype with just hypercalciuria and stone formation.¹⁰⁷

The conversion of circulating serum 25-hydroxyvitamin D ($25[\text{OH}]\text{D}$) to the active hormone calcitriol occurs primarily in the kidneys and is enhanced by vitamin D deficiency.¹⁰⁸ Inactivation is mediated by the enzyme $25(\text{OH})\text{D}$ -24-hydroxylase (CYP24A1), which is also highly expressed in the kidney and converts $1,25(\text{OH})_2\text{D}$ and $25(\text{OH})\text{D}$ to inactive metabolites.¹⁰⁸ Hence CYP24A1 acts as a potent inhibitor of vitamin D action by diminishing both $25(\text{OH})\text{D}$ and $1,25(\text{OH})_2\text{D}$. Biallelic inactivating variants in the gene encoding CYP24A1 (*CYP24A1*) result in elevated serum $1,25(\text{OH})_2\text{D}$, hypercalcemia, and hypercalciuria with nephrolithiasis and nephrocalcinosis.^{109,110} The contribution of monoallelic inactivating variants in *CYP24A1* to the pathogenesis of absorptive hypercalciuria is unclear.

Interestingly, two-thirds of patients with absorptive hypercalciuria exhibit increased intestinal calcium absorption with normal levels of $1,25(\text{OH})_2\text{D}$.^{87,96-99,111} Moreover, a triple-lumen intestinal perfusion study found evidence of $1,25(\text{OH})_2\text{D}$ -independent selective jejunal hyperabsorption of calcium in this population, which is different from the less selective gastrointestinal (GI) effects of $1,25(\text{OH})_2\text{D}$ on normal subjects.¹¹² The pathway and mechanism whereby an increased amount of calcium is absorbed independent of active vitamin D in this population is not known.

Increased Abundance of Vitamin D Receptor

Expression and activation of the vitamin D receptor (VDR) are necessary for vitamin D action.¹¹³ A genetic association of intestinal hyperabsorption of calcium and calcium stone formation with the VDR locus has been described.^{114,115} Two common single-nucleotide polymorphisms (SNPs) were linked to altered expression and/or function of

VDR protein,¹¹⁶⁻¹¹⁹ but these results were not confirmed by others.^{120,121} Because intestinal calcium absorption in hyperabsorptive kidney stone formers mainly occurs with normal serum $1,25(\text{OH})_2\text{D}$,^{87,97-99} there might be an alteration in VDR abundance or sensitivity to $1,25(\text{OH})_2\text{D}$. VDRs are expressed in peripheral monocytes and T and B lymphocytes.¹²²⁻¹²⁶ In 10 male hypercalciuric CaOx stone formers compared with age- and sex-matched controls without a history of stone disease, the abundance of VDR was twofold higher in peripheral blood monocytes of stone formers.¹²⁷ In another study of patients with absorptive hypercalciuria, a Scatchard analysis (a method of linearizing data from a saturation binding experiment to determine binding constants) found increased VDR abundance in a subset of patients with normal serum $1,25(\text{OH})_2\text{D}$ levels. These results suggest heterogeneous patterns with respect to VDR abundance.¹²²

RENAL LEAK HYPERCALCIURIA

The renal leak of calcium is secondary to reduced renal tubular calcium reabsorption and is typically associated with secondary increased serum PTH and $1,25(\text{OH})_2\text{D}$ concentrations and a compensatory increase in intestinal calcium absorption.¹²⁸ The underlying mechanisms mediating impaired renal tubular calcium reabsorption have yet to be fully explored and are a rapidly evolving area of research. Several different mechanisms affect renal tubular calcium reabsorption including a primary proximal defect, hyperinsulinemia, and activating variations in the calcium-sensing receptor (CaSR) claudin-14 axis.¹²⁹ Human metabolic studies demonstrated an exaggerated fractional excretion of calcium (FE_{Ca}) in patients with hypercalciuria in a postprandial state compared with normal subjects.¹³⁰ Importantly, increased FE_{Ca} was not a consequence of differences in filtered calcium load, urinary sodium excretion, or serum PTH concentrations consistent with a primary renal leak (Fig. 40.6).

The CaSR is expressed in multiple tissues including the parathyroid glands, kidney, and GI tract.¹³¹ In the parathyroid glands, elevated plasma calcium activates the CaSR inhibiting PTH release. In the kidney, activation of the CaSR reduces renal tubular calcium reabsorption in the thick ascending limb (TAL) and distal convoluted tubules (DCT).¹³² An Italian cohort has linked hypercalciuria in stone formers to polymorphism in the *CASR* gene. Vitamin D-responsive elements have been identified in the *CASR* gene, which suggests that calcitriol can upregulate kidney CaSR expression, thereby reducing renal tubular calcium reabsorption.¹³³

Alternatively, or in addition to increased renal CaSR expression, genetic variations in the CaSR may lead to its activation at lower plasma calcium concentrations, thereby inducing calciuria. Tight junction proteins of the claudin family expressed in the TAL are critical to paracellular calcium reabsorption in the kidney.¹³⁴ CaSR signaling in the TAL increases claudin-14 expression, which inhibits the reabsorption of calcium through the claudin-16 and claudin-19 complex, thus inhibiting paracellular Ca^{2+} reabsorption in the TAL.¹³⁴ A myriad of genetic association studies are consistent with this mechanism causing hypercalciuria including in patients with nephrolithiasis.^{129,135}

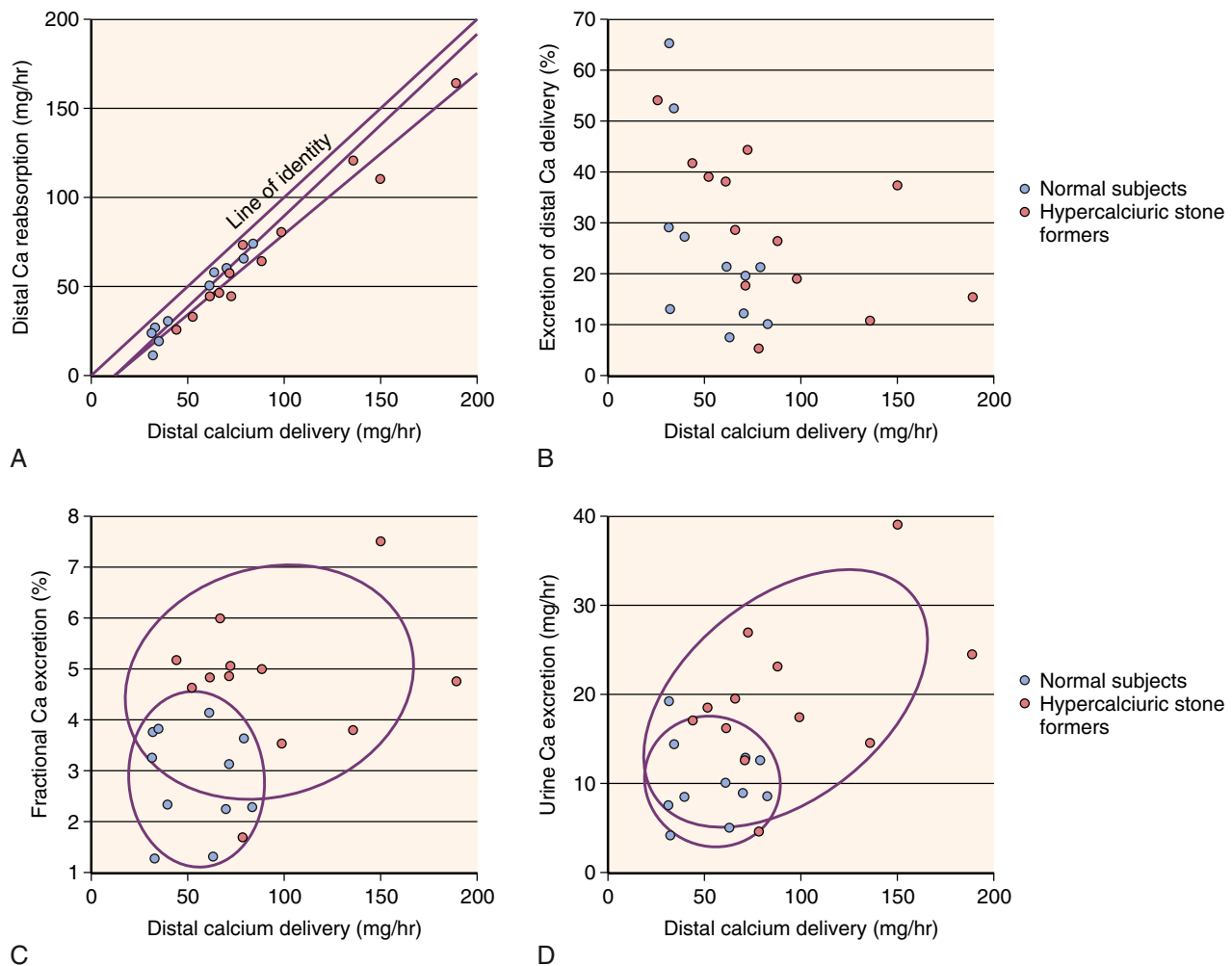


Fig. 40.6 Relationship among distal calcium reabsorption, renal calcium excretion, and distal calcium delivery. (A) Distal calcium reabsorption increases similarly with calcium delivery in normal subjects and hypercalciuric stone formers. Both lines are positioned slightly beneath the line of identity. This suggests normal distal calcium handling but higher distal calcium delivery. (B) The actual fraction (%) of distally delivered calcium that was excreted fell with increasing distal delivery, and stone formers lie above normal at comparable distal delivery. (C) Fractional and total (D) calcium excretions were high in many stone formers versus normal subjects at comparable deliveries. (From Worcester EM, Coe FL, Evan AP, et al. Evidence for increased postprandial distal nephron calcium delivery in hypercalciuric stone-forming patients. *Am J Physiol Renal Physiol*. 2008;295[5]:F1286–F1294.)

RESORPTIVE HYPERCALCIURIA

Resorptive hypercalciuria refers to hypercalciuria caused by enhanced calcium mobilization from bone, which can be PTH dependent or PTH independent. Primary hyperparathyroidism (PHPT) is the most common cause of resorptive hypercalciuria and is associated with calcium stones in 2% to 8% of patients.⁷³ Although hypercalciuria due to increased bone calcium mobilization has been proposed as the cause of kidney stones in this population,^{136,137} the relative contribution of enhanced skeletal calcium mobilization versus enhanced intestinal calcium absorption is unclear.^{136,138,139} Serum 1,25(OH)₂D is higher, and there is an increased calciuric response to an oral calcium load in patients with PHPT and kidney stones, consistent with enhanced intestinal calcium absorption.¹³⁸⁻¹⁴¹

HYPOCITRATURIA

Hypocitraturia occurs in about one-third of patients with calcium nephrolithiasis.¹⁴² Citrate is one of the most abundant organic anions (in molar quantities) in human urine and is an important inhibitor of calcium stone formation via numerous mechanisms: its effects on calcium chelation, prevention of crystallization and aggregation, its effect to reduce ionized calcium, and its ability to promote crystal detachment from cells (Fig. 40.7).¹⁴²⁻¹⁴⁶ Tricarboxylate citrate has pKa values of 2.9, 4.3, and 5.6 and hence is mostly present as a trivalent anion (citrate³⁻) in blood at pH 7.4. Citrate is freely filtered, and approximately 10% to 35% of filtered citrate is excreted in urine, varying with acid-base status. The reabsorption of citrate occurs in the proximal convoluted tubule and, to a lesser extent, in the proximal straight tubule.¹⁴⁷⁻¹⁴⁹ Apical membrane transport occurs through a coupled sodium-dicarboxylate cotransporter (NaDC-1; SLC13A2).^{150,151}

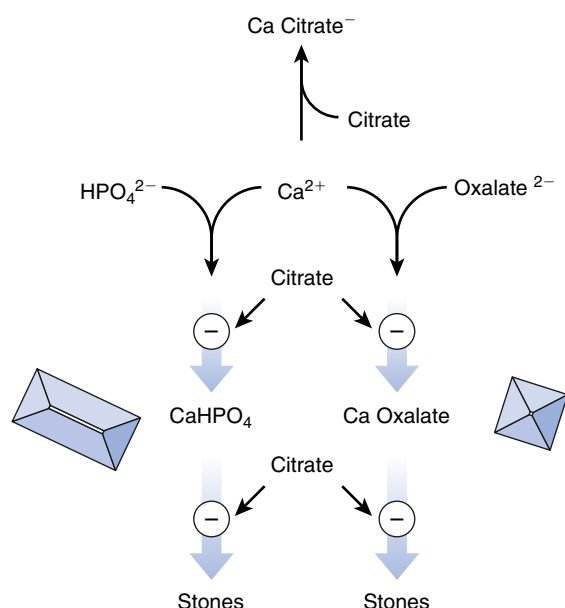


Fig. 40.7 Protective effects of citrate against urolithiasis. Citrate binds calcium with high affinity to form soluble calcium citrate complexes and lowers the ionized calcium and calcium activity. Citrate also directly inhibits calcium oxalate and calcium phosphate crystallization and aggregation.

Role of Acid-Base Factors

Alterations of acid-base status play a key role in renal tubular citrate reabsorption by multiple mechanisms. The highest pK_a of citric acid is 5.6, so the concentration of citrate^{2-} , the transported ionic species, increases at lower luminal pH.¹⁵³ There is also a direct gating effect of pH on NaDC-1 activity, independent of the substrate concentration.¹⁵¹ Acidosis increases the abundance of NaDC-1 in the apical membrane of the proximal tubule. Finally, cytosolic ATP citrate lyase and mitochondrial aconitase activity increase in acidosis, which lowers the intracellular concentration of citrate and promotes its reabsorption.¹⁵⁴⁻¹⁵⁷

Clinical Conditions

Clinical conditions associated with hypocitraturia may be divided into those with systemic extracellular acidosis and those without. Distal RTA is associated with hypocitraturia and is frequently encountered in patients with recurrent nephrolithiasis.¹⁵⁸⁻¹⁶² Other conditions associated with systemic acidosis include carbonic anhydrase inhibition,^{163,164} strenuous physical exercise,¹⁶⁵ and chronic diarrheal states.¹⁶⁶⁻¹⁶⁸

Hypocitraturia is also found in normobicarbonatemic states including incomplete distal RTA, impaired kidney function, mild chronic metabolic acidosis, high protein consumption, thiazide-induced hypokalemia, primary aldosteronism, excessive salt intake, and treatment with angiotensin-converting enzyme (ACE) inhibitors.¹⁶⁹⁻¹⁷⁵

HYPEROXALURIA

Hyperoxaluria is present in isolation or in combination with other risk factors in 8% to 50% of kidney stone formers.^{75,176-178} In CaOx stone formers, both increased urinary oxalate and calcium are responsible for CaOx supersaturation, although the activity of calcium is one

order of magnitude higher than oxalate (Fig. 40.8).¹⁷⁹ Under normal circumstances, the physiologic concentration of the CaOx complex in the urine far exceeds its solubility constant.^{179,180} The mechanisms underlying hyperoxaluria can stem from multiple factors and are summarized in Fig. 40.9.¹⁸¹⁻¹⁸⁵

Increased hepatic production

Oxalate is a dicarboxylic acid and in mammals is either ingested or is an end product of hepatic metabolism.^{74,181,186} Primary hyperoxalurias (PHs) are autosomal-recessive disorders due to enzymatic defects that result in massive hepatic overproduction of oxalate (see also genetic section and Table 40.2). The three genetic forms result in excess glyoxylate, which is the precursor of oxalate. PH1 is the consequence of a deficiency or mistargeting of hepatic alanine-glyoxylate transferase (AGT), which is a pyridoxal 5'-phosphate-dependent enzyme that transaminates glyoxylate to glycine.¹⁸⁷ It accounts for about 80% of PH cases. PH2 is caused by a deficiency in the cytosolic enzyme glyoxylate reductase-hydroxypyruvate reductase (GRHPR), which reduces glyoxylate to glycolate¹⁸⁸ and accounts for about 10% of cases. The least common variety, PH3, is a result of activating mutations in the mitochondrial 4-hydroxy-2-oxoglutarate aldolase enzyme (HOGA), which converts hydroxyproline to glyoxalate.¹⁸⁹⁻¹⁹² PH3 accounts for about 5% of cases; another 5% of PHs have no known genetic mutation.

Dietary Intake and Bioavailability

Dietary oxalate contributes significantly to urinary oxalate excretion. There is a wide variation in estimated oral intake of oxalate, ranging from 50 to 1000 mg/day.^{75,178} Dietary oxalate and its bioavailability contribute approximately 45% of urinary excretion.¹⁹³ The main sources of most bioavailable oxalate-rich food are seeds, chocolate derived from tropical cocoa trees, leafy vegetables, especially spinach and rhubarb, and tea. The relationship between oxalate absorption and dietary oxalate intake has been demonstrated to be nonlinear.¹⁹³

Intestinal Absorption

Despite seminal advances in rodent intestinal physiology of oxalate handling,¹⁹⁴ the specific intestinal segments in the human intestine involved in oxalate absorption and secretion is not fully known. It was proposed that the main site of oxalate absorption is the small intestine because most oxalate absorption occurs in the first 4 to 8 hours after ingestion.^{176,184,195,196} Nevertheless, it has been suggested that the colon may also participate, though to a lesser extent.¹⁹⁶ Oxalate is absorbed and secreted in the GI tract through both paracellular and transcellular pathways.

Role of *Oxalobacter Formigenes*

Many gut bacteria including *Oxalobacter formigenes* (OF) degrade oxalate in the intestinal lumen via oxalate decarboxylase.¹⁹⁷ OF was initially found in ruminates but is also present in other species including humans.^{198,199} Colonization with this bacteria begins during childhood and can be found in feces of 60% to 80% of adults.²⁰⁰ Dietary oxalate intake influences the colonization of the GI tract. In animal studies, a significant decline in urinary

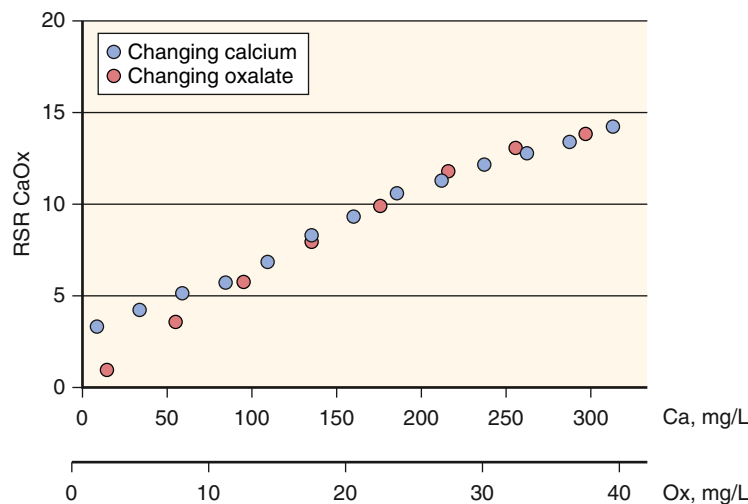


Fig. 40.8 Relationship between calcium oxalate relative supersaturation (RSR) and calcium or oxalate concentration. Calcium or oxalate was varied individually; the other was kept constant. (Modified from Pak CY, Adams-Huet B, Poindexter JR, et al. Rapid communication: relative effect of urinary calcium and oxalate on saturation of calcium oxalate. *Kidney Int.* 2004;66[5]:2032–2037.)

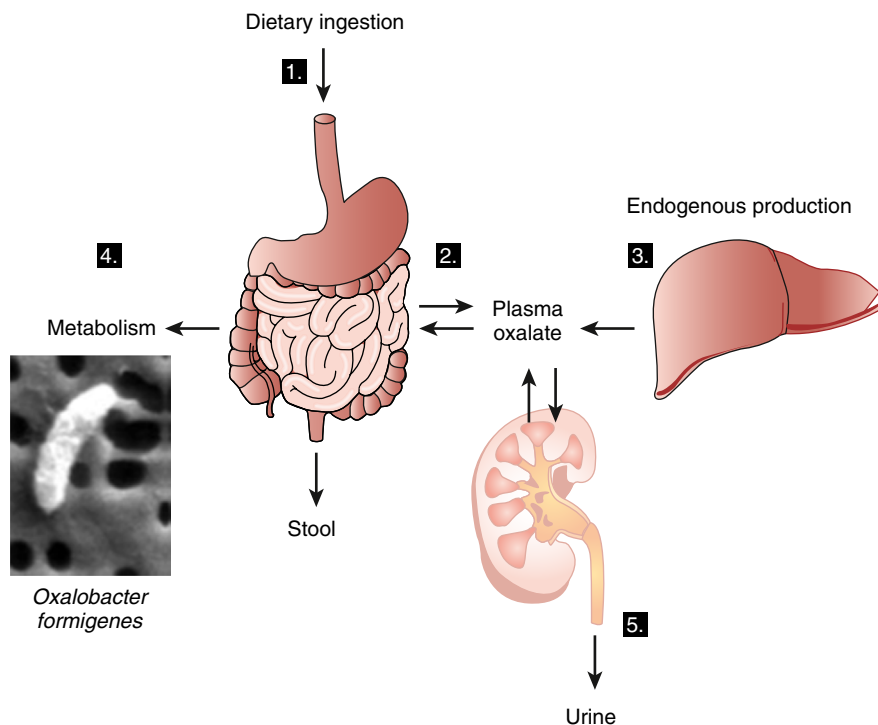


Fig. 40.9 Pathophysiologic mechanisms of hyperoxaluria. Oxalate balance is determined by ingestion and endogenous production versus intestinal and urinary excretion. Intestinal handling can be bidirectional, and luminal degradation is facilitated by microbial degradation. Urinary excretion is the net result of filtration minus tubular reabsorption. Hyperoxaluria can be caused by the following: 1. increased dietary ingestion; 2. increased gut absorption or decreased secretion; 3. increased endogenous hepatic production; 4. decreased intestinal bacterial metabolism; or 5. renal hyperexcretion.

oxalate follows administration of or an increase in OF colonization.^{201,202} In a cross-sectional study, patients with recurrent calcium oxalate stones had lower OF colonization than normal subjects matched for age and sex.²⁰³ However, urinary oxalate excretion was not different between the two populations likely because the study participants were not fed a controlled diet. In another study, Seiner and colleagues²⁰⁴ have shown that OF-negative calcium oxalate

stone formers have higher urinary oxalate excretion than OF-positive patients on a controlled diet.²⁰⁴ Intestinal oxalate absorption, measured using $^{13}\text{C}_2$ -labeled oxalate, was similar in OF-positive and OF-negative patients, but plasma oxalate concentrations were significantly higher in the OF-negative population. The results supported the role of reduced intestinal oxalate secretion in OF-negative calcium oxalate kidney stone formers. In support of this, an ex vivo Ussing

chamber study found that OF degrades intestinal luminal oxalate and also has the capacity to stimulate net intestinal oxalate secretion.²⁰⁵

Renal Excretion

Levels of oxalate in the plasma of healthy individuals approximate 1.5 to 3.0 micromoles per liter. The oxalate/creatinine clearance ratio in healthy subjects is >1 , at first glance indicating a net tubular secretion of oxalate. Moreover, studies with exogenous radiolabeled oxalate in normal humans have shown that renal excretion accounts for most of the disposal of oxalate.²⁰⁶ Oxalate is not significantly protein bound and is freely filtered at the glomerulus. With impaired kidney function, plasma oxalate concentration steadily increases and exceeds its saturation in the blood, therefore enhancing the risk of systemic tissue oxalate deposition. Renal oxalate clearance studies in human subjects are more controversial. Radiolabeled oxalate studies^{207,208} have demonstrated net oxalate secretion. In contrast, endogenous renal oxalate clearance assessments^{209,210} using direct measurements of serum and urine oxalate concentrations observed net reabsorption. However, renal tubular secretion of oxalate leading to an elevated urinary fractional excretion of oxalate has been reported in patients with primary or enteric hyperoxaluria.²⁰⁸⁻²¹⁰

In the renal proximal tubule, both reabsorption and secretion of oxalate have been observed.^{211,212} *Slc26a6* is expressed in the apical membrane of the proximal renal tubule and modulates the activity of several other apical anion exchangers.^{213,214} However, the physiologic role of this anion exchanger in renal oxalate handling has not yet been fully elucidated. In the *Slc26a6* null mice, hyperoxaluria appears to be driven mostly by hyperoxalemia rather than by a renal leak of oxalate.²¹⁵

Clinical Hyperoxaluria

PH1 typically presents during early childhood with calcium oxalate stones and nephrocalcinosis, but the age at onset of symptoms ranges from infancy to the sixth decade. PH1 is the most severe form, with nearly all patients progressing to end-stage renal disease.²¹⁶ Patients with PH2 display a less severe phenotype than those with PH1, yet data indicate that up to 60% of PH2 patients reach end-stage renal disease at the age of 50. PH3 is less severe than PH1 and PH2 and tends to present with symptomatic nephrolithiasis in early childhood but can remit with age. Patients with PH usually display massive hyperoxaluria (>80 mg/day) and pure or predominant calcium oxalate monohydrate stones.

Enteric hyperoxaluria because of inflammatory bowel disease, jejunioileal bypass, and modern bariatric surgeries for morbid obesity is the most common cause of hyperoxaluria in clinical practice.²¹⁷⁻²²² Roux-en-Y gastric bypass is a weight reduction procedure theoretically combining restrictive and malabsorptive mechanisms for the treatment of obesity. Nephrolithiasis is a well-known complication of patients who underwent Roux-en-Y gastric bypass. A comparison in 4690 patients following RYGB and an obese control group has shown 7.5% of kidney stones in patients following Roux-en-Y gastric bypass compared with 4.6% in obese controls. The result of this study was confirmed by another cross-sectional study of 762 patients following bariatric surgery

(mostly Roux-en-Y gastric bypass) matched with obese controls who did not undergo surgery.²²³ Even though the prevalence of kidney stones was similar between the two populations at baseline, following Roux-en-Y gastric bypass, the incidence of nephrolithiasis increased to 11% compared with 4% at follow-up in obese non-stone formers. Another retrospective cohort with 972 patients following Roux-en-Y gastric bypass has shown an 8.8% stone prevalence before surgery, whereas 3.2% developed new stones postoperatively.²²⁴ The pathophysiologic mechanisms for lithogenesis are complex and variable and may be because of hyperoxaluria, hypocitraturia, aciduria, or low urinary volume.^{167,168,225-233}

The underlying mechanisms for hyperoxaluria following inflammatory bowel disease and bariatric procedures are not yet fully defined. Purported mechanisms have linked hyperoxaluria to intestinal fat malabsorption.^{230,234} In this model, unabsorbed fatty acids sequester calcium, which would otherwise bind oxalate in the intestinal lumen, and thereby increase the free luminal concentration of oxalate, enhancing its availability for absorption. An additional mechanism may involve increased permeability of the colon from exposure to unconjugated bile acids and long-chain fatty acids in both inflammatory bowel disease and bariatric procedures.^{185,235} (Fig. 40.10).

ALTERATIONS OF URINARY PH

Both acidic (≤ 5.5) and alkaline (≥ 6.7) urine increase the propensity for calcium kidney stone formation. With unduly acidic urinary pH, urine becomes supersaturated with undissociated UA that can contribute to CaOx crystallization.^{65,236} Highly alkaline urine increases the abundance of monohydrogen phosphate (dissociation constant $pK_a \approx 6.7$), which, in combination with calcium, transforms to thermodynamically unstable brushite ($\text{CaHPO}_4 \cdot 2\text{H}_2\text{O}$) and, finally, to hydroxyapatite, $\text{Ca}_{10}(\text{PO}_4)_6(\text{OH})_2$.

Over the past 4 decades, the average CaP content of stones has progressively increased.²³⁷ The rise in the prevalence of CaP stones has been attributed to, but not proven to be, the result of increasing use of medications including topiramate and alkali therapy.^{39,76,238,239} The three main risk factors for the development of CaP stones are alkaline urine, hypercalciuria, and hypocitraturia. It has been suggested that elevated urinary pH plays the most important role in the transformation of CaOx to CaP stones. A retrospective study of 62 patients has found that high urinary pH is the primary physiologic abnormality in those who evolved from CaOx to CaP.²³⁷ Defective renal acidification was proposed to be related to renal tissue damage from extracorporeal shock wave lithotripsy.³⁹ Patients with CaP stones usually carry a greater stone burden and are less likely to be stone free after urologic procedures; they also experience resistance to extracorporeal shock wave lithotripsy and ultrasonic lithotripsy.²⁴⁰⁻²⁴²

HYPERURICOSURIC CALCIUM UROLITHIASIS

Calcium nephrolithiasis with concomitant hyperuricosuria, also known as hyperuricosuric calcium urolithiasis (HUCU), denotes a peculiar subgroup of calcium stone formers. Increased dietary intake of purine-rich food has been

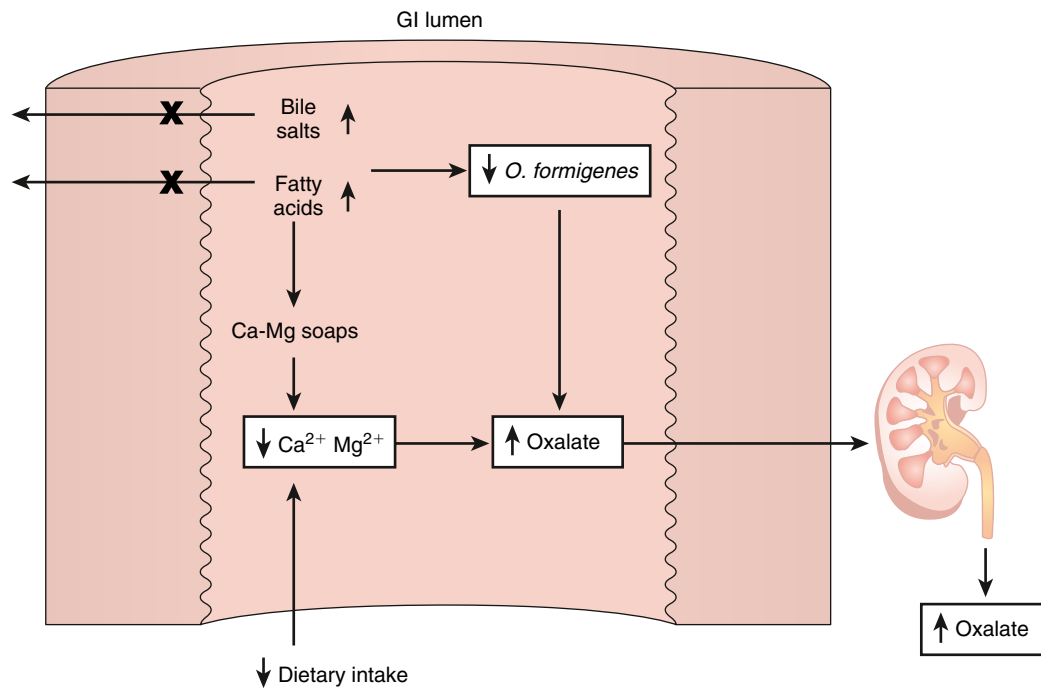


Fig. 40.10 Pathophysiologic mechanisms of hyperoxaluria in inflammatory bowel disease or following bariatric surgery. Fatty acids and bile salts precipitate luminal calcium, which can be compounded by low dietary intake. Low luminal calcium frees up and increases unbound oxalate. Lack of *Oxalobacter formigenes* because of excess bile salts also contributes to high luminal oxalate levels. *GI*, Gastrointestinal.

implicated as the cause of hyperuricosuria in most patients with HUCU, but endogenous overproduction of UA was observed in approximately one-third of HUCU stone formers consuming purine-free diets.^{243,244} The physicochemical basis for hyperuricosuria-induced calcium oxalate stones has been postulated but not yet proven. There are three potential and nonmutually exclusive mechanisms.^{65,245,246} In two studies, the underlying mechanism linking UA to CaOx crystallization was attributed to heterogeneous nucleation^{65,66} (Fig. 40.11). Another study showed that colloidal monosodium urate from supersaturated urine in these patients may attenuate inhibitor activity against CaOx crystallization,^{247,248} but when glycosaminoglycans were specifically examined, UA did not affect its inhibitor activity.²⁴⁹ Monosodium urate can diminish the solubility of CaOx in solution, a process referred to as *salting out*.^{245,250} Despite the uncertainty regarding the mechanism, clinical studies in hyperuricosuric patients have shown a drastic decline in the rate of recurrent kidney stone formation in those treated with xanthine oxidase inhibition.²⁵¹ Contrary to laboratory and clinical evidence, a retrospective population-based study in a large number of patients did not show a relationship between urinary UA and CaOx stone formation.²⁵² The urate effect is likely imperceptible in a large, unselected, calcium stone-forming population.

URIC ACID STONES

The cause of UA nephrolithiasis can be genetic²⁵³⁻²⁵⁵ or acquired,^{165,172,256-259} with metabolic syndrome being the principal cause (Table 40.1). The primary pathophysiologic mechanisms accounting for UA nephrolithiasis include hyperuricosuria and low urinary volume; the most important contributor is an unduly low urinary pH.

PHYSICOCHEMISTRY OF URIC ACID

Given that UA solubility in a urinary environment is limited to about 96 mg/L and UA excretion in humans typically exceeds 600 mg/day, the risk of UA precipitation is constant.^{74,176} UA has a pK_a of 5.35 at 37°C,^{260,261} so UA solubility is determined primarily by pH in that acidic urine ($\text{pH} \leq 5.5$) titrates urate to UA, which is sparingly soluble and therefore precipitates.^{176,262,263} This may also indirectly induce mixed UA-CaOx nephrolithiasis, which can be mediated through heterogeneous nucleation and epitaxial crystal growth.^{67,264,265} Generally, urine is metastably supersaturated with respect to UA. This suggests that the absence of an inhibitor may influence the propensity for UA nephrolithiasis, which is supported by experimental evidence demonstrating the presence of macromolecules that attenuate UA crystal adherence to renal tubular epithelium.²⁶⁶

PATHOPHYSIOLOGY OF URIC ACID STONES

Hyperuricosuria

The principal sources of UA production are de novo synthesis, tissue catabolism, and dietary purine load.²⁶⁷ Generally, 50% of the typical daily urate load is provided by de novo synthesis and tissue catabolism, with the remainder originating from dietary purines.²⁶⁷ The purine bases are guanine and adenine. About 30% of synthesized UA is excreted through the intestinal tract, where it undergoes uricolysis, and the remaining 70% by the kidney.²⁶⁸ Hyperuricosuria may be caused by genetic, metabolic, or dietary factors.¹⁵ Hyperuricosuria is seen in rare hereditary disorders characterized by mutations in enzymatic pathways responsible for UA production, such as X-linked phosphoribosyl pyrophosphate synthetase superactivity

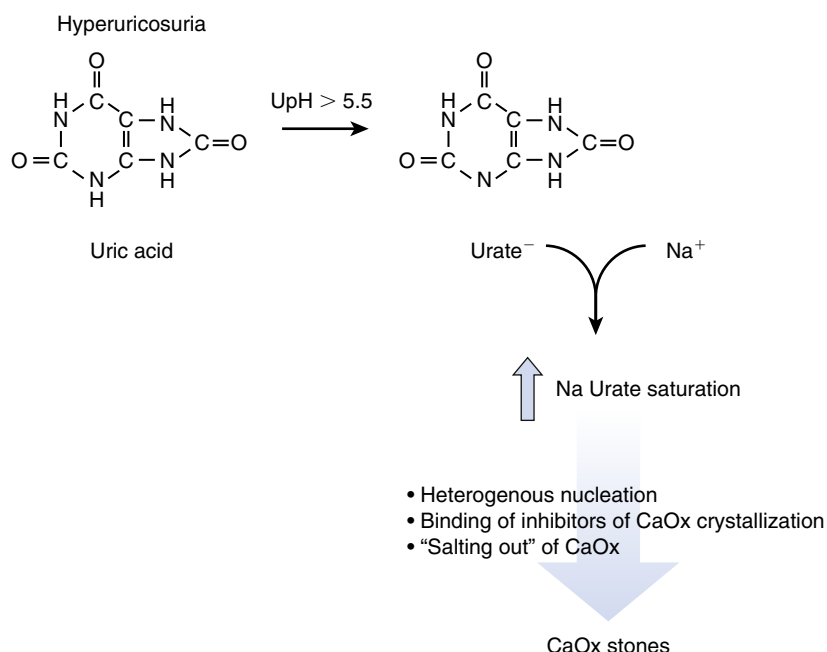


Fig. 40.11 Physicochemical scheme for urate-induced CaOx stones. Hyperuricosuria in the absence of acidic urine renders high free urate activity levels in the urine. Sodium activity is perpetually higher than urate by two orders of magnitude. The higher sodium urate promotes calcium oxalate crystallization by three nonexclusive mechanisms. *UpH*, Urinary pH.

Table 40.1 Causes and Mechanisms for Uric Acid Nephrolithiasis

Causative Factors	Low Urine Volume	Low Urinary pH	Hyperuricosuria
Acquired Conditions			
Diarrhea	+	+	+
Myeloproliferative			+
High animal protein		+	+
Uricosuric drugs			+
Primary gout		+	+
Metabolic syndrome		+	
Monogenic Disease			
Enzyme disorders of purine metabolism			+
Pathogenic variants in uric acid transporters			+

caused by pathogenic variants in *PRPS1* or X-linked Lesch-Nyhan syndrome caused by pathogenic variants in *HPRT1*, encoding hypoxanthine phosphoribosyltransferase (see also genetics section and Table 40.2). Hyperuricosuria may also occur with excessive tissue breakdown in malignancy, especially during chemotherapy.^{269,270} Certain uricosuric drugs such as high-dose salicylates, probenecid, radiocontrast agents, and losartan are known to increase UA excretion, which may increase the risk of UA stone

formation.^{261,271} In the large majority of cases of UA stones, there are no identifiable congenital or acquired causes of hyperuricosuria.

Low Urinary Volume

Low urinary volume increases the urinary supersaturation of all stone-forming constituents.²⁷² Volume depletion from chronic diarrhea in inflammatory bowel disease has an impact on UA stone formation. In addition, UA stones comprise up to two-thirds of all stones in patients following ileostomy.²⁷³ In this population, low urinary volume, in addition to aciduria from intestinal alkali loss, is common.²⁷⁴⁻²⁷⁶

Low Urinary pH

In the majority of cases, hyperuricosuria or low urine volume is not the culprit. Rather, lithogenesis is driven by a low urine pH. This is collectively called, by default, idiopathic UA nephrolithiasis (IUAN).^{176,277,278} IUAN is the stone type that shares numerous common characteristics with the metabolic syndrome.^{258,279-281} Several cross-sectional studies have supported the association among UA stones, diabetes, and obesity²⁸²⁻²⁸⁸ (Fig. 40.12).

Increased acid load to the kidneys and impaired ammonium (NH_4^+) excretion have been shown to be the two main causes of unduly acidic urine in IUAN.^{176,277} Several metabolic studies of a constant controlled metabolic diet have shown that net acid excretion is higher in patients with idiopathic UA nephrolithiasis and type 2 diabetes without stones compared with control subjects,^{277,280,281,289-291} suggesting that endogenous acid production is elevated. The nature and source of these putative organic acids are unknown at present but likely have an enterohepatic origin.²⁹²

In idiopathic UA nephrolithiasis, there is defective NH_4^+ production and excretion by the proximal tubule. As a consequence, more H^+ is buffered by titratable acids

Table 40.2 Mendelian Forms of Nephrolithiasis

Gene	Protein	Disease	Mode of Inheritance	Phenotype	Treatment
AGXT	Alanine-glyoxylate aminotransferase	Primary hyperoxaluria type 1	Autosomal-recessive	Hyperoxaluria (>80 mg or >1000 $\mu\text{mol}/24\text{h}$, calcium oxalate monohydrate stones	Hydration and alkali therapy, pyridoxine, lumasiran/nedosiran, combined kidney/liver transplantation
APRT	Adenine phosphoribosyltransferase	Dihydroxyadeninuria	Autosomal-recessive	2,8-dihydroxyadenine stones, CKD, may recur in kidney transplant	Xanthine oxidase inhibitors, purine restriction, monitor for CKD, radiolucent stones
ATP6V1B1	ATPase, H ⁺ transporting, lysosomal, V1 subunit B1	Distal renal tubular acidosis $\pm$ sensorineural deafness	Autosomal-recessive, incomplete dRTA in some heterozygotes	Metabolic acidosis, alkaline urine pH, CaP stones, nephrocalcinosis	Alkali therapy, audiometry, monitor for CKD
ATP6V0A4	ATPase, H ⁺ transporting, lysosomal, V0 subunit α 4	Distal renal tubular acidosis $\pm$ sensorineural deafness	Autosomal-recessive	Metabolic acidosis, alkaline urine pH, CaP stones, nephrocalcinosis	Alkali therapy, audiometry, monitor for CKD
CA2	Carbonic anhydrase II	Proximal and distal renal tubular acidosis	Autosomal-recessive	Metabolic acidosis, alkaline urine pH, nephrocalcinosis, osteopetrosis	Alkali therapy
CASR ^a	Calcium-sensing receptor (activating mutations)	Autosomal-dominant hypocalcemia, Bartter syndrome type 5	Autosomal-dominant	Hypocalcemia, hypercalciuria, normal PTH, calcium stones	Avoid calcium and vitamin D, calcilytics
CASR ^a	Calcium-sensing receptor (inactivating mutations)	Familial hypocalciuric hypercalcemia	Autosomal-dominant	Hypercalcemia, hypocalciuria, normal to high PTH, calcium stones	Calcimimetics
CLCN5	Chloride channel, voltage-sensitive 5	Dent disease type 1	X-chromosomal recessive	Low-molecular weight proteinuria, CaP stones, nephrocalcinosis	Monitor for CKD, kidney transplantation
CLCNKB	Chloride channel, voltage-sensitive Kb	Bartter syndrome type 3	Autosomal-recessive	Hypokalemia, metabolic alkalosis, hypercalciuria, CaP stones	Mineral supplementation, NSAIDs
CLDN2	Claudin-2	Calcium nephrolithiasis	X-linked	Hypercalciuria and nephrolithiasis	Treat hypercalciuria
CLDN16	Claudin 16	Familial hypomagnesemia with hypercalciuria and nephrocalcinosis	Autosomal-recessive; evidence suggesting incomplete dominance	Renal calcium and magnesium wasting	Treat hypercalciuria, monitor for seizures
CLDN19	Claudin 19	Familial hypomagnesemia with hypercalciuria and nephrocalcinosis	Autosomal-recessive; evidence suggesting incomplete dominance	Renal calcium and magnesium wasting, ocular abnormalities	Treat hypercalciuria, ophthalmologic screening
CYP24A1 ^a	Cytochrome P450 1,25(OH) ₂ D 24-hydroxylase	Infantile idiopathic hypercalcemia	Autosomal recessive; evidence suggesting incomplete dominance	Hypercalcemia, hypercalciuria, elevated 1,25 Vit D, suppressed PTH	Restrict excess vitamin D and sun exposure, fluconazole

Table 40.2 Mendelian Forms of Nephrolithiasis (Continued)

Gene	Protein	Disease	Mode of Inheritance	Phenotype	Treatment
<i>FAM20A</i>	Family with sequence similarity 20, member A	Enamel renal syndrome	Autosomal-recessive	Nephrocalcinosis, amelogenesis imperfecta	Monitor for CKD
<i>FOXI1</i>	Forkhead box protein I1	Distal renal tubular acidosis ± sensorineural deafness	Autosomal-recessive	Metabolic acidosis, alkaline urine pH, nephrocalcinosis	Alkali therapy, audiometry, monitor for CKD
<i>GRHR</i>	Glyoxylate reductase/hydroxypyruvate reductase	Primary hyperoxaluria type 2	Autosomal-recessive	Hyperoxaluria, calcium oxalate monohydrate stones	Hydration and alkali therapy, kidney transplantation
<i>HNF4A^b</i>	Hepatocyte nuclear factor 4, alpha (p.R76W)	MODY, Fanconi syndrome, nephrocalcinosis	Autosomal-dominant	Fanconi syndrome, hypercalciuria, hypermagnesemia, nephrocalcinosis, CKD	Monitor for CKD, treat hypercalciuria
<i>HOGA1</i>	4-hydroxy-2-oxoglutarate aldolase 1	Primary hyperoxaluria type 3	Autosomal-recessive	Hyperoxaluria, calcium oxalate monohydrate stones	Hydration and alkali therapy
<i>HPRT1</i>	Hypoxanthine phosphoribosyltransferase 1	Lesch-Nyhan-syndrome/Kelly-Seegmiller-syndrome	X-chromosomal recessive	Hyperuricosuria, UA stones, gout, neurologic features	Xanthine oxidase inhibitors, purine restriction
<i>KCNJ1</i>	Potassium inwardly rectifying channel, subfamily J, member 1	Barter syndrome type 2	Autosomal-recessive	Hypokalemia, metabolic alkalosis, hypercalciuria, CaP stones	Mineral supplementation, NSAIDs
<i>OCRL</i>	Inositol polyphosphate 5-phosphatase	Dent disease type 2 (Lowe syndrome) X-chromosomal recessive	X-chromosomal recessive	Low-molecular-weight proteinuria, CaP stones, nephrocalcinosis CKD, kidney transplantation	Monitor for CKD and seizures, kidney transplantation, ophthalmologic screening
<i>PRPS1</i>	Phosphoribosyl pyrophosphate synthetase	Phosphoribosyl pyrophosphate synthetase superactivity	X-chromosomal recessive	Hyperuricemia, UA stones, gout, CKD, neurodevelopmental abnormalities, and sensorineural deafness	Xanthine oxidase inhibitors, purine restriction
<i>SLC2A9</i>	Glut9, Solute carrier family 2 (facilitated glucose transporter), member 9	Renal hypouricemia type 2	Autosomal-recessive	Hyperuricosuria, hypouricemia, UA stones, EAKI	Hydration, alkali, and xanthine oxidase inhibitors to prevent EAKI
<i>SLC3A1</i>	Basic amino acid transporter system b ^{0,+}	Cystinuria (type A)	Autosomal-recessive	Elevated urine cystine, cystine stones	Hydration, protein and sodium restriction, alkalinization therapy, thiol, monitor for CKD
<i>SLC4A1</i>	Solute carrier family 4, anion exchanger, member 1	Distal renal tubular acidosis (sometimes incomplete)	Autosomal-dominant	Metabolic acidosis, alkaline urine pH	Alkali therapy, monitor for CKD

Continued on following page

Table 40.2 Mendelian Forms of Nephrolithiasis (Continued)

Gene	Protein	Disease	Mode of Inheritance	Phenotype	Treatment
SLC7A9	Basic amino acid transporter system b ⁰ ,+	Cystinuria (type B)	Autosomal-recessive, autosomal-dominant with incomplete penetrance	Elevated urine cystine, cystine stones	Hydration, protein and sodium restriction, alkalinization therapy, thiol, monitor for CKD
SLC9A3R1	Solute carrier family 9, subfamily A member 3 regulator 1	Hypophosphatemic nephrolithiasis, osteoporosis-2	Autosomal-dominant	Calcium stones, low bone density	
SLC12A1	Solute carrier family 12, member 1	Bartter syndrome type 1	Autosomal-recessive	Hypokalemia, metabolic alkalosis, hypercalciuria, CaP stones	Mineral supplementation, NSAR
SLC22A12	Urat1, solute carrier family 22 (organic anion/urate transporter), member 12	Renal hypouricemia type 1	Autosomal-recessive	Hyperuricosuria, hypouricemia, UA stones, (EIAKI)	Hydration, alkali, xanthine oxidase inhibitors to prevent EIAKI?
SLC34A1 ^a	Sodium-dependent phosphate cotransporter 2a (NPT2a)	Hypophosphatemic nephrolithiasis, osteoporosis-1/Idiopathic infantile hypercalcemia	Autosomal-recessive; evidence suggesting incomplete dominance	Hypophosphatemia, elevated 1,25(OH)2D, hypercalciuria, nephrocalcinosis, calcium stones	Phosphate supplementation, restrict excess vitamin D, bone density measurements
SLC34A3 ^a	Sodium-dependent phosphate cotransporter 2a (NPT2c)	Hereditary hypophosphatemic rickets with hypercalciuria	Autosomal-recessive; evidence suggesting incomplete dominance	Hypophosphatemia, elevated 1,25(OH)2D, hypercalciuria, nephrocalcinosis, calcium stones	Phosphate supplementation, restrict excess vitamin D, bone density measurements
WDR72 ^a	WD-repeat containing protein 72	Distal renal tubular acidosis	Autosomal-recessive	Metabolic acidosis, alkaline urine pH, nephrocalcinosis, amelogenesis imperfecta	Alkali therapy
XDR	Xanthine dehydrogenase	Xanthinuria type 1	Autosomal-recessive	Xanthine stones, hypouricemia	Purine restriction, hydration, radiolucent stones

^aGenes also found as candidate loci in genome-wide association studies on kidney stone disease.^bFanconi syndrome and nephrocalcinosis have only been observed with a specific allele: p.R76W.

CaP, Calcium phosphate; EIAKI, exercise-induced acute kidney injury; CKD, chronic kidney disease; MODY, maturity onset diabetes of the young; NSAID, nonsteroidal anti-inflammatory drug; PTH, parathyroid hormone; UA, uric acid.

Adapted from Pollicastro LJ, Saggi SJ, Goldfarb DS, Weiss JP. Personalized intervention in monogenic stone formers. *J Urol*. 2018;199(3):623–632 and Halbritter J. Genetics of kidney stone disease—polygenic meets monogenic. *Nephrol Ther*. 2021;17S:S88–S94.

(including urate) and the reduced buffer capacity results in a lower urine pH.^{176,277,281} The tradeoff is a propensity for UA precipitation.^{277,278} Defective NH_4^+ excretion is not unique to UA stone formers but remains a shared feature between metabolic syndrome and type 2 diabetic non-stone formers.^{258,279,281,288} The exact mechanisms of defective NH_4^+ excretion is unknown but may be caused by lipotoxicity due to ectopic fat deposition in the kidney.²⁹³⁻²⁹⁵ Another potential mechanism is attenuated NH_4^+ synthesis in the proximal tubular because of substrate competition—namely, fatty acid instead of glutamine as a source of energy—thus removing the source of nitrogen for ammoniogenesis.²⁹⁶

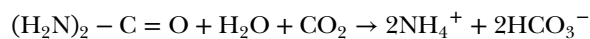
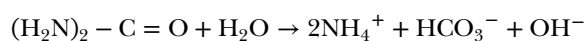
INFECTION-RELATED STONES

PATHOPHYSIOLOGY

This category of stones does not form because of intrinsic host defects but because of an infected urinary environment. Struvite ($\text{MgNH}_4\text{PO}_4 \cdot 6\text{H}_2\text{O}$) stones account for a low percentage of all kidney stones and often also contain carbonate apatite, $\text{Ca}_{10}(\text{PO}_4)_6 \cdot \text{CO}_3$. These stones are rapidly growing, and they branch, enlarge, and fill the renal collecting system to form staghorns. The key culprits are urea-splitting bacteria, especially *Proteus* species. These stones are difficult to treat medically. Even with surgical removal, any remaining fragments containing the infecting bacteria furnish a nidus for further rapid stone growth.

Struvite stones occur more frequently in women than in men, largely because of the higher incidence of urinary tract infections. Chronic urinary stasis or infections predispose to struvite stones so that older age, neurogenic bladder, indwelling urinary catheters, and urinary tract anatomic abnormalities are all predisposing factors. The presence of large stones in infected alkaline urine should alert the clinician to the potential presence of struvite. Given their potential for rapid growth and substantial morbidity, early detection and eradication are essential.^{297,298}

The urease of urea-splitting organisms hydrolyzes the following reactions:



Urea contains two nitrogens and one carbon. On the product (right-hand) side of the equation, 2NH_4^+ represents two acid equivalents, and either HCO_3^- plus OH^- or 2HCO_3^- represents two base equivalents. Phosphate and magnesium combine with NH_4^+ to form struvite, and the calcium and phosphate combine with carbonate to form carbonate apatite. During infections with urease-producing organisms, there is a simultaneous elevation in urine NH_4^+ , pH, and carbonate concentration. With successful antimicrobial treatment of the underlying infection, struvite can dissolve because the urine is generally undersaturated with respect to struvite.^{299,300} However, urine is not undersaturated with respect to carbonate apatite,³⁰⁰ so successful antimicrobial therapy is not expected to dissolve this component of the stone. Whether a stone will dissolve with prolonged antibiotics is dependent on the amount of carbonate apatite.

Although many bacteria (gram-negative and gram-positive), *Mycoplasma*, and yeast species can produce urease, most urease-producing infections are caused by *Proteus mirabilis*. In addition, *Haemophilus*, *Corynebacterium*, and *Ureaplasma* have also been reported to cause struvite stones. All these bacteria use urease to split urea and supply their need for nitrogen (in the form of NH_3).

UNCOMMON ACQUIRED KIDNEY STONES

Rare acquired kidney stones may be iatrogenic due to the use of medications or caused by toxins or other diseases.³⁰¹ Genetic kidney stone disease will be discussed in the next section.

Ammonium urate stones are radiolucent stones that occur in patients with chronic diarrhea, inflammatory bowel disease, ileostomy, and laxative abuse; all are

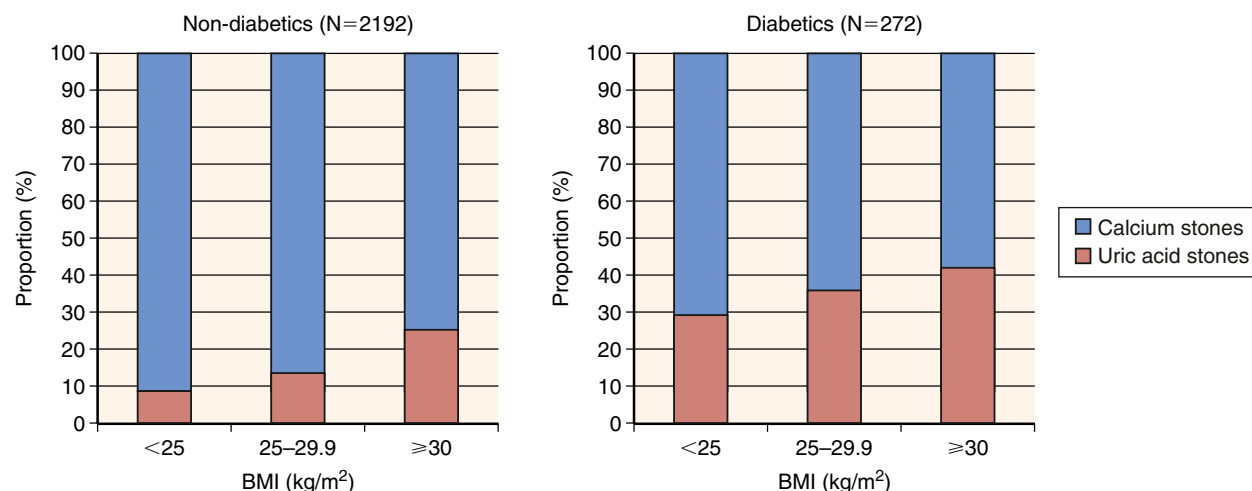


Fig. 40.12 Distribution of stone type with respect to body mass index (BMI) and diabetes status. (Based on data from Daudon M, Traxer O, Conort P, et al. Type 2 diabetes increases the risk of uric acid stones. *J Am Soc Nephrol*. 2006;17[7]:2026–2033.)

associated with intestinal alkali loss and compensatory renal hyperexcretion of ammonium.³⁰²⁻³⁰⁶ This occurs because of the relative abundance of urinary ammonium accompanied with decreased urinary sodium and potassium levels, thereby making the urine supersaturated with respect to poorly soluble ammonium urate. The catabolic state of these patients may also cause hyperuricosuria.

Consumption of over-the-counter drugs, such as guaifenesin as an expectorant or ephedrine as a stimulant, accounts for approximately one-third of drug-induced kidney stones in the United States.^{307,308} After the introduction of protease inhibitors for the treatment of patients with human immunodeficiency virus, indinavir-treated patients began to display a high incidence of indinavir-associated stones.^{309,310} In later years, it was shown that other antiproteases, such as nelfinavir, tenofovir, atazanavir, and antinucleosidic drugs including efavirenz also cause kidney stones.³¹¹⁻³¹⁴ Other drugs such as methotrexate, triamterene, acyclovir, various antimicrobial agents (e.g., sulfonamides, penicillin, cephalosporin, quinolones [especially ciprofloxacin], and nitrofurantoin) and magnesium trisilicate are causes of drug-induced stones.³⁰¹

Other environmental factors play a role in kidney stone formation. Melamine is an organic nitrogenous compound used in the industrial production of plastics, dyes, fertilizers, and fabrics.³¹⁵ In 2008, kidney stone cases were reported in infants and children in China consuming melamine-contaminated milk.³¹⁶ A study in Taiwan screened 1129 children with potential exposure to contaminated milk formula. The researchers found that those with a high exposure had an increased incidence of nephrolithiasis. The age of the group of children with kidney stones was reported to be significantly younger than those without. Metabolic workups did not disclose any evidence of hypercalciuria, and the stones were radiolucent, indicating that these stones were related to melamine ingestion.^{317,318}

GENETICS

Familial clustering of kidney stones has long been recognized.³¹⁹ A positive family history is present in 20% to 50% of individuals with kidney stones,³²⁰⁻³²³ and twin studies³²⁴⁻³²⁷ revealed a strong heritability of nephrolithiasis of 45% in women and 56% in men. A similar strong heritability has been observed for urinary traits associated with nephrolithiasis, even after adjustment for demographic, anthropometric, and dietary covariates.^{325,327,328} The genetics of nephrolithiasis is heterogeneous and complex, and there has been an enormous increase of knowledge in the field of kidney stone genetics in the past 2 decades. The spectrum ranges from rare but highly penetrant variants causing Mendelian disease (monogenic disease) to more common but phenotypically milder variants associated with nephrolithiasis (polygenic traits). Large genome-wide association studies (GWAS) in different ethnic cohorts revealed >20 genetic loci associated with the risk of kidney stones, together accounting for ~14% of the heritability of nephrolithiasis.^{105,329-333} Genes identified by GWAS

are involved in renal water and solute transport, vitamin D metabolism, and calcium-sensing receptor signaling. Interestingly, several risk alleles, identified by GWAS, localize to genes causing Mendelian forms of nephrolithiasis (*CASR*, *CYP24A1*, *SLC34A1*, *SLC34A3*, and *WDR72*), suggesting a continuous spectrum from monogenic to polygenic traits due to intermediate effect size variants (dominant disorders) or genotype dosage effects (recessive disorders). The discovery of these risk alleles provided important mechanistic insights into the pathogenesis of kidney stone disease in humans. Yet genetic screening for these risk alleles remains mainly a research tool and has no established role in the clinical care of individuals with kidney stones.

The situation is different with monogenic kidney stone disease. More than 30 Mendelian forms have been identified thus far with autosomal-dominant, autosomal-recessive, or X-linked transmission (Table 40.2). Interestingly, in several of these disorders, both recessive and dominant modes of inheritance have been reported, suggesting a genotype dosage effect. In small, selected cohorts of mostly early onset or familial nephrolithiasis at tertiary centers, a monogenic cause was identified in 6.8% to 29.4% of cases.³³⁴⁻³³⁷ The prevalence of monogenic disease was higher in pediatric than adult cohorts, and recessive mutations were more frequently observed in children, whereas dominant mutations were more abundant in adults.³³⁴ However, the frequency of monogenic disease in sporadic adult-onset nephrolithiasis, by far the most common type encountered in the clinic, has not been established.

Recognition of monogenic disease is essential for accurate genetic counseling, informed decision making by families and physicians, counseling about the role of kidney transplantation, and access to appropriate medical support and services, as well as to direct patients to clinical trials and registries.³³⁸ Furthermore, the pathophysiology is well defined in many cases, enabling tailored treatments in affected patients. However, the diagnosis of Mendelian forms of nephrolithiasis remains a challenge, the main obstacle being differentiation from idiopathic stone disease. A detailed history and thorough clinical examination of the patient, as well as the alertness of the treating physician, are key. Red flags pointing to inherited kidney stone disease are early onset of disease, positive family history, consanguinity of parents, chronic kidney disease (CKD), nephrocalcinosis, signs of tubular dysfunction (e.g., hypophosphatemia and low-grade proteinuria), or extrarenal manifestations.

Monogenic causes of nephrolithiasis are listed in Table 40.2. A detailed description of all Mendelian nephrolithiasis forms is beyond the scope of this chapter. The reader is referred to excellent reviews on the topic.³³⁹⁻³⁴² In this chapter, we cover a selection of monogenic diseases with known pathophysiology and well-established therapeutic interventions.

CYSTINURIA

Cystinuria is the most common monogenic kidney stone disease and caused by biallelic pathogenic variants, deletions, or duplications in the *SLC3A1* or *SLC7A9* genes,

which encode for the subunits rBAT or b^{0,+}AT, respectively, of a heterodimeric dibasic amino acid transporter in the proximal tubule (Fig. 40.13). The rBAT and b^{0,+}AT subunits are linked by a disulfide bridge in the heterodimer, which is characteristic of the heteromeric amino acid transporters³⁴³ and mediates obligatory exchange of cationic amino acids (cysteine, arginine, ornithine, lysine) and neutral amino acids with 1:1 stoichiometry.

There are no systemic symptoms of amino acid deficiency in this disorder, but it causes 1% to 2% of renal stones in adults and 4% to 8% in pediatric patients.^{344,345} Normally, reabsorption of amino acids from the urine is almost complete. Inactivation of rBAT/b^{0,+}AT leads to urinary wasting of several cationic amino acids. These amino acids have good solubility, but cysteine can dimerize to form cystine (cystine = two cysteines bound by a disulfide bridge) that has poor water solubility. Because rBAT/b^{0,+}AT is also present in the gut, there is some malabsorption of cationic amino acids, but there is no clinically discernible intestinal phenotype.

GENETICS

Many missense/nonsense and splice-site variants have been described in the causative genes *SLC3A1* and *SLC2A9*, respectively. Gross deletions and insertions are common; hence genetic testing should include methods enabling detection of copy number variation, such as next-generation sequencing or multiplex ligation-dependent probe amplification (MLPA).³⁴⁵ Comprehensive mutation analysis in both genes yields a molecular diagnosis in up to 85% of patients, but a significant fraction of patients with clinical cystinuria remains without a genetic diagnosis.³⁴⁶ This may be due to missed pathogenic variants in the promoter, regulatory, or intronic regions or the presence of additional cystinuria genes. *SLC7A13* has been proposed as a novel cystinuria-associated gene, but there are currently no data supporting a role of pathogenic *SLC7A13* variants in cystinuria.^{346,347} Patients presenting with monoallelic *SLC7A9* variants and calcium rather than cystine stones have been reported.^{348,349} It remains unclear at the moment if this

reflects just a coincidence or if there is a pathophysiologic basis for this association. Cystine has been shown to promote growth and aggregation of calcium oxalate crystals in human urine.³⁵⁰ Hence higher-than-normal urine cystine in these heterozygous carriers may promote formation of calcium stones.

CLINICAL PRESENTATION

The worldwide prevalence is estimated to be 1:7000 with high ethnogeographic variation.³⁴⁵ Patients present with hematuria, renal colic, and urinary obstruction, although their symptoms tend to be more severe and they are more likely to have staghorns, surgery, and CKD.³⁵¹ The age of onset varies widely but probably 50% of patients present in childhood.³⁵² The historical type I versus type II classification has been replaced by type A (biallelic mutations in *SLC3A1*; genotype AA) and type B (biallelic mutations in *SLC7A9*; genotype BB).³⁵³ Digenic (*SLC3A1* and *SLC7A9*) inheritance (type AB) has also been described, but individuals with this pattern are less likely to suffer from kidney stones because they have a 25% chance of having a completely wild-type b^{0,+} system. Neither the phenotypic nor genotypic classification appears to correlate with the clinical course of affected patients. Genetic testing is not mandatory for the diagnosis, yet it may be useful for genetic counseling, situations of clinical uncertainty, or in cases with a prenatal diagnosis of hyperechogenic colon.

The features that should raise suspicion for cystinuria in stone formers are consanguinity of parents, early disease onset, large or recurrent stones, concomitant CKD, positive nonquantitative screening test with sodium nitroprusside (>75 mg/L; 0.325 mM), and the presence of pathognomonic hexagonal cystine crystals on urinalysis (see Fig. 40.13).

Quantitation of urinary cystine excretion (normal, <30 mg/day; <0.13 mmol/day) is mandatory to diagnose cystinuria. Patients with cystinuria often excrete >400 mg/day (>1.7 mmol/day). Specific assays for cystine have been developed to distinguish between cystine and cystine-thiol drug complexes.³⁵⁴

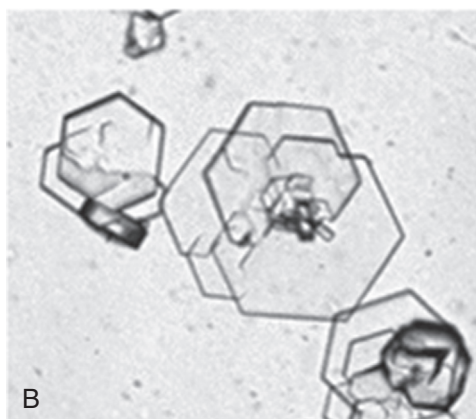
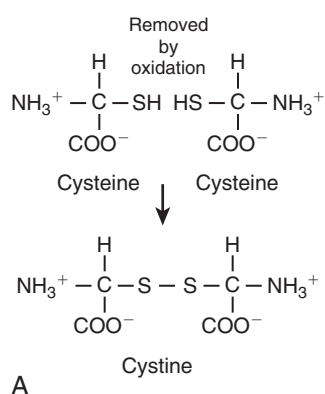


Fig. 40.13 (A) Cystine is a dimer of cysteine formed by disulfide bonding under oxidizing conditions. (B) Appearance of cystine crystals in the urine of a patient with cystinuria.

MANAGEMENT

There are no randomized trials evaluating the efficacy of dietary or pharmacologic interventions in patients with cystinuria. Fluid, alkali, and dietary modification (salt and protein) are first-line therapies, with thiol therapy as second line. To reduce urine cystine supersaturation to <1.0 or concentration <243 mg/L (1 mmol/L) requires drinking around 4 L/day of fluids including nocturnal intake.³⁵⁴ Cystine excretion has been shown to be reduced by decreasing dietary sodium, but the physiology of this effect is not known.³⁵⁵ Reduction of animal protein has also been proposed because it decreases the intake of methionine, the precursor of cysteine.³⁵⁶

Pharmacologic Therapy

Cystine solubility increases with increasing pH, which can be achieved with oral potassium citrate. One can start with 20 mEq three times daily. The goal is to achieve a urinary pH >7 . Sodium citrate and sodium bicarbonate should be avoided because the sodium load can increase urine cystine and calcium excretion. In $>50\%$ of patients, the above outlined measures do not result in a sustained reduction in stone recurrence and treatment with a thiol agent is indicated. The two available agents are D-penicillamine and α -mercaptopyropionylglycine, or tiopronin. These drugs work by reducing the disulfide bond of cystine, producing mixed compounds with cysteine, which are more soluble than cystine.³⁵⁷ The incidence of adverse effects is lower with tiopronin, thus rendering it as the preferred first-line agent. D-Penicillamine is often used for patients with adverse reactions to tiopronin. The prescription of these drugs is discussed in more detail in a subsequent section.

ADENINE PHOSPHORIBOSYLTRANSFERASE DEFICIENCY

Adenine phosphoribosyltransferase (APRT) deficiency, also known as 2,8-dihydroxyadeninuria, is a rare autosomal-recessive disorder caused by biallelic pathogenic variants in *APRT*. APRT is a purine salvage enzyme that converts adenine to adenosine monophosphate. APRT deficiency results in excessive production and urinary excretion of 2,8-dihydroxyadenine (DHA) and formation of DHA stones. With APRT deficiency, adenine is converted to 8-hydroxyadenine, which is further metabolized to DHA by xanthine dehydrogenase.³⁵⁸ Consequently, APRT deficiency results in high urinary levels of DHA, which is insoluble and forms crystals that aggregate, grow, and ultimately form kidney stones.^{359,360} Patients may also develop CKD secondary to crystal precipitation in the renal parenchyma. The disease is often underdiagnosed and can recur in kidney transplants. APRT deficiency can present at any age, but in approximately 50% of subjects, symptoms do not occur until adulthood.³⁶¹ DHA crystals are pathognomonic for the disease. The crystals are reddish-brown on hematoxylin-eosin and periodic acid-Schiff stains, black on silver stain, blue on trichrome stain, and birefringent under polarized light. With polarized microscopy, DHA crystals in the urine are round and reddish-brown, with a characteristic central Maltese cross

pattern.³⁵⁸ Prognosis is excellent with allopurinol therapy started before CKD ensues, resulting in soluble adenine instead of insoluble 2,8-DHA crystals.

NEPHROLITHIASIS AS A SYSTEMIC DISORDER

Traditionally, kidney stone disease has been recognized as an isolated, benign, painful local condition of the urinary tract. However, this is not the case. The association of nephrolithiasis with gout and degenerative vascular disease in postmortem examinations was noted by Morgagni in the 1760s.³⁶² In recent years, the prevalence of kidney stones has increased, along with the ever-expanding epidemic of obesity, type 2 diabetes mellitus, and metabolic syndrome.³⁶³⁻³⁶⁹ Concern with metabolic syndrome and the risk for nephrolithiasis has not been limited to adults because, as discussed earlier, this link has also been reported in overweight and obese adolescents and children with kidney stones.^{16,370} It is presently unclear whether the link between kidney stone disease and metabolic syndrome reflects the same underlying pathophysiologic mechanisms in both disorders or is simply an association.

OBESITY, WEIGHT GAIN, DIABETES MELLITUS, AND RISK FOR NEPHROLITHIASIS

Obesity and weight gain increase the risk of kidney stones³⁶³ (Fig. 40.14). The relative risk for stone formation in men with a body weight of 100 kg or more is significantly higher than men with a body weight 68 kg or less.³⁶³ Similarly, the relationship between T2DM and the risk of kidney stone formation has been clearly demonstrated in three large cohorts including the Nurses' Health Study I comprising older women, the Nurses' Health Study II consisting of younger women, and the Health Professionals Follow-up

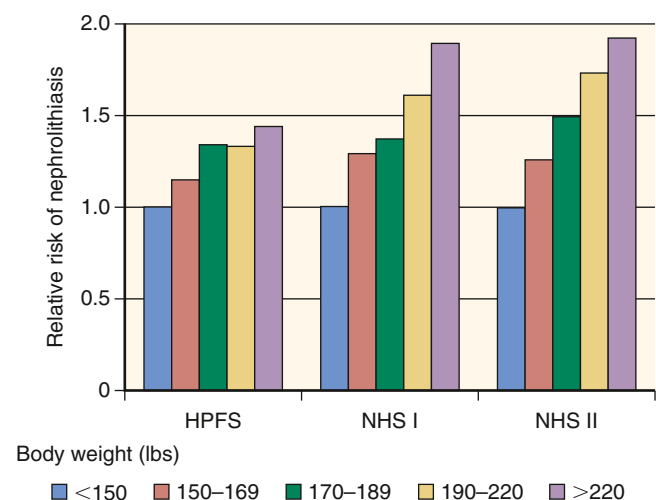


Fig. 40.14 Obesity and the risk of nephrolithiasis. The data were pooled from three databases. HPFS, Health Professionals Follow-Up Study; NHS I, Nurses' Health Study I; NHS II, Nurses' Health Study II. (Modified from Sakhaee K, Maalouf NM, Sinnott B. Clinical review. Kidney stones 2012: pathogenesis, diagnosis, and management. *J Clin Endocrinol Metab.* 2012;97[6]:1847–1860.)

Study in men. The relative risk of prevalent kidney stones in persons with T2DM compared with those without is 1.38 in older women, 1.60 in younger women, and 1.31 in men.³⁶⁴

ASSOCIATION BETWEEN METABOLIC SYNDROME AND NEPHROLITHIASIS

Metabolic syndrome is characterized by a number of features including dyslipidemia, hyperglycemia, hypertension, obesity, and insulin resistance.^{371,372} In addition to its relations to T2DM and cardiovascular risks, metabolic syndrome is associated with nephrolithiasis and CKD.^{16,363-376} Despite the observed association between metabolic syndrome and kidney stone disease, it remains unclear if one stone type or a subset is driving this. The link between obesity and CaOx nephrolithiasis has been in part explained by dietary factors, such as a higher consumption of salt and animal protein.^{377,378} The contribution of metabolic syndrome to urinary calcium and urinary oxalate excretion in calcium stone formers has not been fully elucidated.³⁷⁹ Uric acid stones are more prevalent in patients with T2DM and kidney stone disease than in stone formers without diabetes and among those who are obese relative to those who are nonobese.^{277,282-285,287} Higher Quetelet (body mass) index (BMI) and T2DM are independent risk factors for UA nephrolithiasis. Total body and truncal fat associate with risk factors for UA stone formation.³⁸⁰

NEPHROLITHIASIS, CARDIOVASCULAR DISEASE, AND HYPERTENSION

Traditional Framingham risk factors for coronary artery disease such as atherosclerosis, hypertension, diabetes, and metabolic syndrome occur frequently in patients with kidney stones.^{364,365,381-385} Several observational studies support an increased risk of coronary artery disease in patients with kidney stone disease; however, some have failed to do so.³⁸⁶⁻³⁸⁹ There may be varying risks of cardiovascular disease in males versus female kidney stone formers explaining these discrepant results³⁹⁰ (Fig. 40.15).³⁹¹ Importantly, a causal relationship among kidney stones, coronary artery disease, and sex has not clearly been demonstrated.

There is a clear association between nephrolithiasis and blood pressure, with kidney stone formers more likely to have hypertension.³⁹²⁻³⁹⁶ The causal relationship has not yet been fully demonstrated, but some have suggested that alterations in calcium metabolism may potentially link the development of kidney stone disease and hypertension.^{397,398} Hypercalciuria, which is prevalent in subjects with calcium nephrolithiasis and primary hypertension, has been proposed as a major underlying mechanism, but the role of calcium in human hypertension has not been established.³⁹⁷⁻⁴⁰¹

NEPHROLITHIASIS AND CHRONIC KIDNEY DISEASE

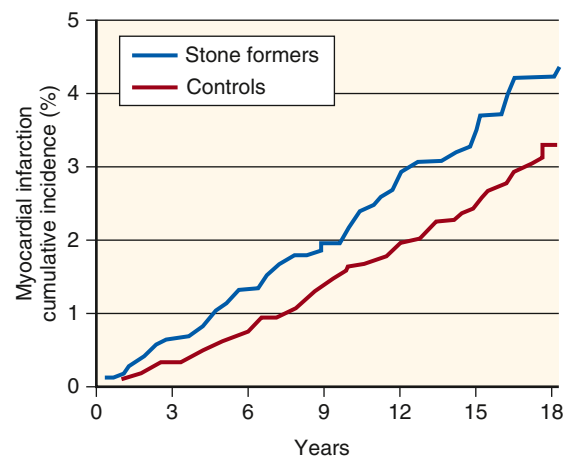
Kidney stone formers experience an increased risk of CKD and ESKD.^{402,403} Kidney stone disease and CKD can potentially be causally linked through recurrent obstruction and infection, through repeated shock wave therapy, or to common comorbid conditions. The prevalence of ESKD attributed to nephrolithiasis in the general population

has been estimated to be approximately 3.1 cases/million per year or approximately 1%.^{404,405} Interestingly, the association between kidney stones and CKD is sex specific, with a significantly higher risk for ESKD, doubling of serum creatinine level, and CKD (stages 3b–5) in women with a history of kidney stones than in men.⁴⁰³

CALCIUM STONES AND BONE DISEASE

EPIDEMIOLOGY

Bone disease is an underemphasized condition in nephrolithiasis. Several studies reported lower bone mineral density (BMD) and a higher risk of fracture in individuals with a history of kidney stones compared with individuals without.^{83,406-408,409,410} (Fig. 40.16). In a study with nearly 10,000 women with a self-reported history of kidney stones followed for more than 8 years, however, no independent association between nephrolithiasis and incident fracture was found.⁴¹¹ These conflicting results may be explained by confounding factors in population-based cohort studies, such as diet, supplements, and medications known to affect both the risk of kidney stones and bone disease or lack of adjustment for differences in ethnicity and health status.^{48,406,411-414} In an attempt to overcome some of these limitations, one study investigated the association of nephrolithiasis with incident wrist and hip fractures in the Nurses' Health Study ($N = 107,001$ women; 32 years of follow-up) and the Health Professionals Follow-up Study ($N = 50,982$ men; 26 years of follow-up). After exclusion of fractures due to major trauma and multivariable adjustment for age, designated race, BMI, medication, supplements, and diet, nephrolithiasis was found to be independently associated with a higher risk of incident wrist fractures (relative risk 1.20, 95% confidence interval [CI] 1.08–1.33) but not hip fractures (relative



Incidence (no. at risk)

Control	0 (10,860)	0.8 (6,689)	2 (3,184)	4.2 (1,010)
Stone formers	0 (4,564)	1.3 (2,686)	3 (1,276)	5.2 (404)

Fig. 40.15 Increased risk for myocardial infarction in stone formers. Data collected from Olmsted County, Minnesota, residents. (Modified from Rule AD, Roger VL, Melton LJ 3rd, et al. Kidney stones associate with increased risk for myocardial infarction. *J Am Soc Nephrol.* 2010;21[10]:1641–1644.)

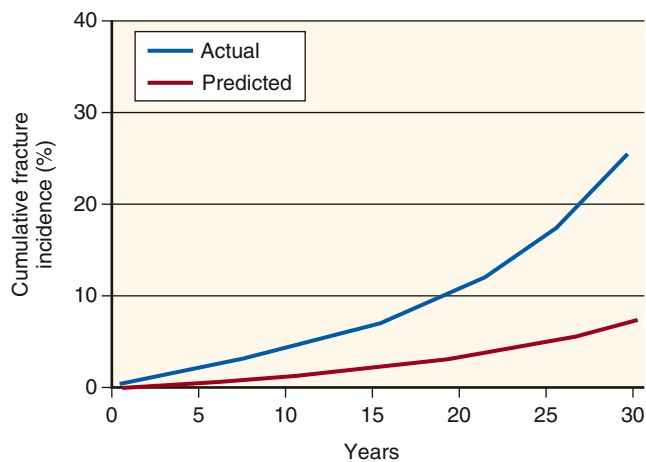


Fig. 40.16 Cumulative incidence of vertebral fractures in stone formers; data from Rochester, Minnesota, residents following an initial episode of symptomatic nephrolithiasis. The elevated fracture risk was vertebral and was present in both sexes. (Modified from Melton LJ 3rd, Crowson CS, Khosla S, et al. Fracture risk among patients with urolithiasis: a population-based cohort study. *Kidney Int.* 1998;53[2]:459–464.)

risk 0.94, 95% CI 0.82–1.08). Thus only a fraction of the excess risk of fractures in kidney stone formers seems to be independently associated with nephrolithiasis. Comorbidities, medication, diet, and other factors associated with nephrolithiasis therefore contribute to increased fracture risk in kidney stone formers.

PATHOPHYSIOLOGIC MECHANISMS LINKING BONE DISEASE AND KIDNEY STONES

Low BMD is common in calcium kidney stone formers.^{83,103,415–443} Loss of BMD is seen at all skeletal sites, with 40% of patients showing diminished BMD at the vertebral spine, 30% at the proximal hip, and 65% at the radius.⁸³ Although low BMD is present in hypercalciuric and normocalciuric stone formers,^{417,420,424} it is most prominent in those with hypercalciuria.^{417,420,424,429,435} Diminished BMD was not universally detected in normocalciuric kidney stone formers.^{423,426,429,431}

Most histomorphometric studies have agreed that defective bone formation rather than excessive bone resorption plays a key role in the development of bone disease in this population.^{103,436–439,441,444} (Table 40.3). The underlying pathophysiologic mechanism(s) are unknown. It has been assumed that the negative calcium balance associated with idiopathic hypercalciuria is a key factor for bone loss in kidney stone formers.^{423,445–447} Yet defective bone formation would not be expected if only hypercalciuria played a pathogenetic role in the development of bone disease in this population.^{103,441} Idiopathic hypercalciuria is a heterogeneous and complex disorder that could result from increased production or sensitivity to calcitriol and affect the target organs differently. Thus while there is evidence of increased calcitriol-mediated intestinal calcium absorption in this population, the high levels of calcitriol commonly encountered in hypercalciuric stone formers may result in attenuated osteoblastic bone formation.^{100,127,448–452}

Dietary factors likely play an important role in the development of bone disease in kidney stone formers.

Salt and protein intake are associated with an increased risk of kidney stones⁸³ and bone disease.^{423,424} Multiple pathophysiologic mechanisms contribute to the high salt- and protein-induced hypercalciuria including diminished renal tubular calcium reabsorption and buffering of dietary acid in bone.^{173,453–458} Subclinical metabolic acidosis is common in protein-induced hypercalciuria.⁴⁵⁶ Metabolic acidosis inhibits osteoblastic matrix protein synthesis and alkaline phosphatase activity.^{459–461} Low bicarbonate in vitro alters osteoblastic extracellular matrix proteins including type I collagen^{459,461} osteopontin, matrix Gla protein,^{460,462} and expression of cyclooxygenase-2⁴⁶³ and RANKL.^{464,465}

There is also evidence that genetic factors contribute to bone disease in kidney stone formers. A genome-wide association study revealed that common sequence variants in the claudin-14 gene (*CLDN14*) are associated with kidney stones and reduced BMD at the hip.³³⁰ In a larger follow-up study, variants in *ALPL* (encoding alkaline phosphatase), *CASR* (encoding the calcium-sensing receptor), and *SLC34A1* (encoding the sodium/phosphate cotransporter 2a) were also found to be associated with kidney stones and BMD.³²⁹ In addition, several of these variants were also associated with mineral metabolism parameters including alkaline phosphatase (*ALPL* variants), parathyroid hormone (*SLC34A1* and *CLDN14* variants), serum phosphate (*SLC34A1* variants), and serum magnesium (*CLDN14* variants). *CASR* gene variants were found to associate with urinary calcium in kidney stone formers⁴⁶⁶ and reduced forearm BMD in healthy subjects and postmenopausal women.^{467,468} Thus variants in these genes may affect bone turnover directly (*ALPL*: alkaline phosphatase activity in osteoblasts) or indirectly through modulation of mineral metabolism (*SLC34A1*, *CLDN14*, and *CASR*).

EVALUATION

CLINICAL PRESENTATION

SYMPTOMS AND SIGNS

Renal colic, pain localized to the back and flank during stone passage, is a common clinical manifestation of kidney stones, although the majority of patients may remain asymptomatic for a long time. The pain occurs as the stone is propelled through the ureter and is a consequence of increased intraluminal pressure, causing stimulation of nerve endings in the ureteral mucosa. Pain is usually intense and intermittent, originating in the back or flank, radiating around the torso to the groin, and ending up in the testicles or labia for male or female patients, respectively. Stones in the midportion of the ureter may imitate appendicitis on the right side or diverticulitis on the left side. Renal colic can be associated with symptoms such as nausea and vomiting because the GI tract shares common innervation with the genitourinary system. When the stone approaches the urinary bladder, it frequently causes symptoms such as urinary frequency, dysuria, suprapubic pain, and incontinence.⁴⁶⁹ The abdominal examination is usually negative. Younger children in particular may not display these typical symptoms and instead may have nausea and

Table 40.3 Bone Histomorphometric Characteristics in Kidney Stone Formers

Reference	Study Participants	No. of Participants	Bone Histomorphometric Profiles
Bordier ¹⁰³	Dietary hypercalciuria	20	None
	Renal hypercalciuria	19	Increased osteoclast and osteoblast surfaces
	Hypophosphatemia	21	Increased osteoclastic and eroded surface (within normal range), decreased osteoblast surfaces, decreased osteoid parameters
Malluche et al., 1977, ⁴³⁶	Controls	12	Low-normal osteoclastic bone resorption, low osteoblastic activity, decreased fraction of mineralizing osteoid seams, decreased mineralization apposition rate
	Absorptive hypercalciuria	15	
de Vernejoul ⁴³⁷	Controls	22	Decreased trabecular volume, decreased active osteoblastic surface, decreased active bone resorption surface
	Idiopathic hypercalciuria	30 (20 M, 10 F)	
Steiniche ⁴³⁸	Controls	187	Increased bone resorption surfaces (decreased refilling of lacunae with low bone formation), decreased bone formation rate, increased mineralization lag times
	Idiopathic hypercalciuria	33 (22 M, 11 F)	
Heilberg ⁴³⁹	Controls	30 (19 M, 11 F)	Increased eroded surface, decreased osteoid surface, decreased bone formation rate with a complete lack of tetracycline double labeling
	Fasting hypercalciuria	6 M	
Bataille ⁴⁴⁰	Controls	No information	Low eroded surface, low bone volume, low osteoid surface, thickness, mineral apposition rate, adjusted apposition rate and bone formation rate
	Idiopathic hypercalciuria	24 (20 M, 4 F)	
Misael da Silva ⁴³¹	Controls	18 (9 M, 9 F)	High eroded surface, increased osteoblastic bone surface, no change in trabecular thickness
	Idiopathic hypercalciuria	22	
Heller ⁴⁴¹	Control	94	Relatively high bone resorption (osteoclast surface, bone surface—mean value within the normal limit), lower indices of bone formation (osteoblast surface/bone surface), decreased wall thickness
	Absorptive hypercalciuria	9 (6 M, 3 F)	
	Controls	9 (6 M, 3 F)	

vomiting or diffuse abdominal pain; consequently, a higher index of suspicion is required.⁴⁷⁰

MEDICAL HISTORY

Clinical evaluation of kidney stone formers should always include a comprehensive medical history including dietary and fluid consumption records and a review of prescribed and over-the-counter medication and supplements.

Kidney stone disease is associated with multiple comorbidities, so patients should be evaluated for conditions such as arterial hypertension, diabetes mellitus, metabolic syndrome,^{364,471,472} recurrent urinary tract infections, inflammatory bowel disease, bowel resection, pancreatic disease, bariatric surgery, and medullary sponge kidney.^{74,473,474} Disorders of calcium homeostasis, such as primary hyperparathyroidism and conditions accompanied by extrarenal calcitriol production (e.g., granulomatous diseases such as sarcoidosis), may also directly contribute to kidney stone disease.⁴⁷⁵ Besides, patients with nephrolithiasis are at higher risk of systemic complications such as bone disease, CKD, cardiovascular events, and vascular calcifications.⁴⁷⁶⁻⁴⁷⁹

Environment and lifestyle habits, such as prolonged exposition to higher temperatures and heavy physical activity,

can also increase the risk of kidney stones, if fluid losses are not adequately replenished.^{474,480,481} A careful medication history is critical because some drugs have been demonstrated to increase the risk of kidney stones,⁴⁷³ either indirectly by altering the urinary environment (vitamin C supplements, carbonic anhydrase inhibitors, laxatives,^{302,303} probenecid, topiramate, lipase inhibitors,⁴⁸²⁻⁴⁸⁴ and chemotherapeutic agents⁴⁷³), or directly, due to drug-crystallization into the urine because of their poor solubility (triamterene, protease inhibitors, guaifenesin, ephedrine, antacids, and antimicrobials such as sulfonamides and quinolones).^{164,239,473,474,485}

Red flags pointing to inherited kidney stone disease are early onset of disease, positive family history, consanguinity of parents, CKD, nephrocalcinosis, tubular dysfunction, or extrarenal manifestations.

LABORATORY WORKUP

Laboratory workup in kidney stone formers consists of a panel of blood and 24-hour urine chemistry and stone composition analysis (Table 40.4),^{74,469} aimed at separating idiopathic from secondary and genetic forms of nephrolithiasis, identification, and quantification of the primary urinary proliothogenic abnormalities and a

Table 40.4 Ambulatory Metabolic Evaluation and Interpretation of Urinary Parameters^a

Random 24-hour Urinary Profile	Expected Values (Per Day)	Interpretation
Total volume	≥2.5 L	Indicative of daily fluid intake (minus insensible losses); diminishes with low fluid intake, sweating, and diarrhea
pH	5.9–6.2	<5.5—increases risk of uric acid precipitation; commonly found in idiopathic uric acid stone patients, individuals with intestinal disease and diarrhea, and in those with intestinal bypass surgery >6.7—commonly found in patients with dRTA, primary hyperparathyroidism, alkali, and carbonic anhydrase treatment >7.0–7.5—indicates urinary tract infection from urease-producing bacteria
Sodium	100 mmol	Reflects dietary sodium intake (minus extrarenal loss); much lower than dietary intake in diarrhea and with excessive sweating; high sodium intake is major cause of hypercalciuria
Potassium	40–60 mmol	Reflects dietary potassium intake (minus extrarenal loss); much lower than dietary intake in diarrhea states; gauge of dietary alkali intake because most dietary potassium accompanied by organic anions
Calcium	≤250–300 mg (≤6.24–7.49 mmol)	A higher value expected in males; in states of zero balance, urinary calcium excretion is net gut absorption minus net bone deposition; secondary causes should be ruled out before making the diagnosis of idiopathic hypercalciuria
Magnesium	30–120 mg (1.23–4.94 mmol)	Low urinary magnesium detected with low magnesium intake, intestinal malabsorption (small bowel disease), and following bariatric surgery; low magnesium may increase risk of calcium stones
Oxalate	≤45 mg (≤0.51 mmol)	High urinary oxalate is commonly encountered with intestinal disease with fat malabsorption, such as inflammatory bowel disease and following bariatric surgery; values >100 mg/day (1.14 mmol/day) suggest primary hyperoxaluria (PH), the definitive diagnosis should be made by genetic testing
Phosphorus	≤1100 mg (35.5 mmol)	Indicative of dietary organic and inorganic phosphorus intake and absorption; a higher excretion may increase the risk of calcium phosphate stone formation
Uric acid	600–800 mg (3.57–4.76 mmol)	Hyperuricosuria is encountered with overproduction of endogenous uric acid or overindulgence of purine-rich foods such as red meat, poultry, and fish; mainly a risk factor for calcium oxalate stones when UpH is >5.5 but is a risk factor for uric acid stones when UpH <5.5.
Sulfate	≤20 mmol	Sulfate is a marker of dietary acid intake (oxidation of sulfur-containing amino acids, present mainly in animal-derived protein).
Citrate	≥320 mg (≥1.67 mmol)	Inhibitor of calcium stone formation; hypocitraturia is commonly encountered in metabolic acidosis, dRTA, chronic diarrhea, excessive protein ingestion, strenuous physical exercise, hypokalemia, intracellular acidosis, with carbonic anhydrase inhibitor drugs (e.g., acetazolamide, topiramate, and zonisamide), but rarely with ACE inhibitors
Ammonium	30–40 mmol	Ammonium is a major carrier of H ⁺ in the urine; its excretion corresponds with urinary sulfate (acid load); a higher ammonium-to-sulfate ratio indicates GI alkali loss.
Chloride	100 mmol	Chloride varies with sodium intake.
Cystine	<30–60 mg (<0.12–0.25 mmol)	Cystine has a limited urinary solubility, ≈243 mg/L (=1 mmol/L at a urine pH of 7)

^aExpected values should be cross-checked with reference laboratory recommendations because these values may differ. These limits are mean +2 standard deviations (for calcium, oxalate, uric acid, pH, sodium, sulfate, and phosphorus) or mean –2 standard deviations (for citrate, pH, and magnesium) from normal.

ACE, Angiotensin-converting enzyme; dRTA, distal renal tubular acidosis; GI, gastrointestinal; UpH, urinary pH.

search for comorbid conditions that are associated with nephrolithiasis. This is of particular importance in children who should all be investigated after one episode.

SERUM CHEMISTRY

All kidney stone formers require the determination of full fasting serum chemistries (e.g., electrolytes, including

calcium and phosphorus, acid-base status, renal function, and UA) and PTH. Fasting glucose and a full lipid panel are also justified considering the high prevalence of diabetes and metabolic syndrome in stone formers. The measurement of serum 1,25(OH)₂D may be considered only for specific situations such as sarcoidosis or suspicion of inactivating mutations in the 1,25(OH)₂D-24-hydroxylase

gene (*CYP24A1*), which regulates the degradation of $1,25(\text{OH})_2\text{D}$ in $24,25(\text{OH})_2\text{D}$ and occurs more commonly in children. A serum $25(\text{OH})\text{D}$ measurement is helpful in patients with high or high-normal PTH levels and mild hypercalcemia to exclude vitamin D deficiency as a cause of the high PTH level.^{74,469,474,486}

SPOT URINALYSIS

Although a 24-hour urine collection is considered the gold standard, in some situations a spot urine analysis may be performed. Obtaining a 24-hour urine specimen can be impractical in certain patients, such as younger children. One study suggested that an afternoon urine collection in children could serve as an alternative to a 24-hour collection.⁴⁸⁷ In adults, there is poor agreement between spot and 24-hour urine analysis results.⁴⁸⁸ This also applies to urine pH: a fasting urine pH measurement cannot be used as a substitute for 24-hour urine pH.⁴⁸⁹ Urine pH measurements by dipstick are notoriously inaccurate. However, a very low dipstick urine pH (<5.5) suggests UA stones, whereas a high urine pH (>6.5) suggests dRTA and a very high pH (>7.5) should raise suspicion for infection.

CRYSTALLURIA

The analysis of urine crystals can provide useful and complementary information to a 24-hour urine analysis. It can support making a diagnosis (e.g., cystinuria and drug crystals), especially in the absence of a stone composition analysis. In addition, it may also predict the risk of stone recurrence. When repeatedly observed in a patient, crystalluria is highly predictive of the risk of stone recurrence.^{488,490} However, crystalluria per se is not abnormal because crystals can be found in urines of healthy individuals, except for specific types, which are never present in normal urine (e.g., cystine crystals). The first-morning void is the best sample for the assessment of crystalluria, yet collection of the first-morning void is often impractical. Therefore the second morning urine is commonly used.⁴⁹¹ Crystals are typically assessed qualitatively (crystal type) and quantified by polarized light microscopy in the high-power field (400× magnification).^{488,492}

24-HOUR URINARY METABOLIC WORKUP

Although not without flaws, studies have highlighted a significant correlation between kidney stone composition and urinary biochemical profiles.^{493,494} Therefore recurrent kidney stone formers and first-time stone formers with risk factors should be informed of the possibility of performing a 24-hour urine investigation for prevention and treatment purposes.^{488,495,496} The following should be considered as additional risk factors, warranting a metabolic workup in adults: 1. first stone event occurring <25 years of age and a positive family history for nephrolithiasis; 2. secondary conditions associated with an increased risk of stone formation (see medical history, environmental and lifestyle risk factors); 3. CKD with eGFR <60 mL/min/1.73 m²; 4. presence of bilateral or multiple stones; 5. stone events in patients with a solitary kidney or those who are kidney transplant recipients; 6. noncalcium stone composition; and 7. concurrent nephrocalcinosis or osteoporosis. Patients with a single stone event without risk factors may be considered

at low risk of recurrence, so a metabolic evaluation is not mandatory and general dietary advice can be given to reduce the risk of recurrence and avoid complications such as osteopenia and CKD. In contrast, all children with kidney stones should undergo metabolic workup.

The 24-hour urinary metabolic workup aims to estimate the tendency for urinary crystallization and to investigate on the metabolic mechanism underlying nephrolithiasis, determining the risk of CKD and extrarenal involvement. Because up to 45% of patients may show significant differences between two 24-hour urine samples, even if collected within a short time span, at least two 24-hour urine collections (on a random outpatient diet) are recommended to capture the full spectrum of metabolic abnormalities.⁴⁹⁷ This is a key clinical point. Patient education is critical for proper urine collection to both motivate the patient to complete the collections properly and ensure that the collections represent the patient's normal lifestyle and dietary practices.⁴⁹⁸⁻⁵⁰⁰

Regardless of the number of samples collected, physicians should take into account the dietary consumption of known risk factors such as salt and animal protein intake, which can be estimated from 24-hour urinary sodium and urea nitrogen excretion.⁵⁰¹ A complete 24-hour urine profile should include the measurements of urine volume and urinary solutes that can influence the propensity of urine to form crystals, such as sodium, potassium, calcium, phosphate, magnesium, chloride, urea, UA, citrate, and oxalate. Importantly, storage collection conditions for individual components may not be compatible, necessitating distinct collections for individual components, though this is laboratory dependent. To establish urine collection adequacy, the 24-hour urine creatinine excretion can be measured. Creatinine excretion is determined by many variables, especially by body weight and composition, with population reference ranges varying between 13 and 29 mg/kg/24 h in men and between 9 and 26 mg/kg/24 h in women. At a given body weight, more muscle mass results in larger degrees of creatinine generation; in contrast, sarcopenia, with or without obesity, typically results in lower degrees of creatinine generation. For persons of average or near-average body composition, lower values may suggest an inadequately collected sample, whereas higher values may indicate overcollection.⁵⁰² Examination of the reported urine volumes can help here. Improperly collected 24-hour urine samples or incorrect preanalytical handling of urine samples may lead to inadequate treatment recommendations.

Ultimately, a comprehensive metabolic workup may also include a BMD analysis, given the high prevalence of low bone mass and increased fracture risk in kidney stone formers compared with the general population.⁸³

URINARY SUPERSATURATION ESTIMATION

Urinary supersaturation for lithogenic salts can be used in clinical and research settings to estimate the risk of recurrence and monitor response to treatment (Table 40.5).^{493,494,503} The most widely used program to calculate supersaturations is the EQUIL2 software.^{44,504,505} Other programs including JESS and LithoRisk have also been used.^{51,506} Studies have shown a strong correlation

Table 40.5 Urinary Supersaturation Estimation^a

Term	Definition	Readout
Relative supersaturation ratio (RSR)	Ion concentration product in a given solution (activity product); ion concentration product at equilibrium (solubility product)	Values <1 represent undersaturation; values >1 represent supersaturation
Relative supersaturation (RS)	Urinary activity product in urine specimen; activity product from normal subjects	Upper normal limits = 2

^aThe EQUIL2 software program is used for the assessment of urinary supersaturation in stone research, as well as in a clinical setting. The JESS program, which is now mostly used in research programs, also represents the supersaturation ratio but is defined as saturation index in that software program.

between stone type and the prevailing supersaturations in 24-hour urine samples calculated by EQUIL2.^{49,494} A post-hoc analysis of a randomized controlled trial (RCT) for dietary stone prevention demonstrated that a 20% reduction in urinary supersaturation for CaOx at 1 week was associated with a 16% reduction in the risk of stone recurrence.⁵⁰

STONE COMPOSITION ANALYSIS

Stone composition analysis provides critical information that clarifies the differential diagnosis and thereby complements the metabolic workup.^{507,508} Stone analysis also assists in the diagnosis of rare stones, such as cystine, 2,8-hydroxyadenine, or drug-induced stones.^{473,474} Hence a stone analysis should be obtained in all kidney stone formers at least once. In recurrent stone formers who are unresponsive to medical treatment, repeat stone composition analysis should be performed as stone type or composition may have changed. A major limitation of stone composition analysis has been the lack of a standard method among different laboratory techniques to identify urinary stone composition.^{509,510} However, consensus guidelines now strongly suggest that chemical analysis of urinary stone composition must be substituted with more accurate spectroscopy techniques including Fourier-transform infrared spectroscopy or X-ray diffraction.⁴⁸⁸

IMAGING STUDIES

Imaging studies should be considered in patients suspected of having kidney stones and in the follow-up of treated stone formers to monitor stone activity. Various recognized imaging methods include plain abdominal radiography for the kidneys, ureters, and bladder (KUB); ultrasound

examination; digital tomosynthesis; and non-contrast-enhanced computed tomography.

KIDNEYS, URETERS, AND BLADDER RADIOGRAPHY

KUB is a plain radiograph of the abdomen with the advantage of wide availability, minimal radiation exposure, and low cost. The major drawback is limited sensitivity (45%–58%) and specificity (60%–77%).⁵¹¹ However, KUB may provide useful information on stone phenotype because calcium oxalate, calcium phosphate, struvite, and cystine are typically radiopaque, while pure UA stones are radiolucent.⁵¹²

ULTRASOUND

Ultrasound is a reliable, noninvasive, and rapid technique and does not require ionizing radiation. The major limitation is its low sensitivity. Average sensitivity and specificity of 45% and 94%, respectively, for detection of ureteral stones and 45% and 88%, respectively, for kidney stones have been reported.⁵¹² The lack of radiation and contrast renders it safe, particularly for children and pregnant women and is thus the first choice for these patient populations.^{513,514} It is also inexpensive and widely available. It is excellent in detecting hydronephrosis and hydroureter⁵¹⁵ and suitable for patients with radiolucent stones (e.g., UA). However, ultrasound may miss a significant fraction of ureteral stones⁵¹⁵ and may give a false-positive diagnosis of obstruction in patients with pyelonephritis, vesicoureteric reflux, and residual dilation after relief of obstruction. Finally, sonography tends to overestimate the size of a stone because of the inaccurate determination of the stone and tissue boundary.

COMPUTED TOMOGRAPHY

Non-contrast-enhanced computed tomography has the highest sensitivity (94%–100%) and specificity (92%–99%)^{516,517} and can be considered the gold standard for diagnosis of kidney stones.⁵¹⁸ Stones as small as 1 mm can be diagnosed. The disadvantages of this technique include radiation exposure, limited ability to evaluate degree of obstruction, and high cost. Newer techniques that offer lower exposure to radiation (from traditional 8–16 millisieverts [mSv] down to 0.5–2 mSv) have been adapted.⁵¹⁹ Another advantage of the computed tomography (CT) scan is the ability to determine stone density using Hounsfield units (HU), which yields information on stone composition (UA stones typically display <500 HU and calcium-containing stones >500 HU) and may have prognostic value in terms of success in shock wave therapy.⁵²⁰ Similarly, as a preoperative test (e.g., before percutaneous nephrolithotomy), non-contrast-enhanced CT can detect possible altered anatomy and accurately assess stone size and location, both of which have an impact on the selection of optimal surgical intervention.

INTRAVENOUS PYELOGRAPHY

Although intravenous pyelography (IVP) has historically been the gold standard imaging approach for diagnosing nephrolithiasis, it has now been largely supplanted by non-contrast-enhanced CT. IVP might still occasionally be used after non-contrast-enhanced CT in some patients to

guide percutaneous or endoureteral surgical procedures or to confirm a medullary sponge kidney diagnosis in doubtful cases.⁵²¹

MAGNETIC RESONANCE IMAGING

Magnetic resonance imaging (MRI) is a potential alternative to NCCT for the diagnosis of nephrolithiasis and urinary tract obstruction. The obvious advantage is that it does not deliver ionizing radiation; therefore it is useful in pregnant women, especially when US is nondiagnostic. Intravenous gadolinium contrast is typically not administered and should be avoided entirely in the pregnant patient. Using standard MRI sequences, stones appear as a nonspecific signal void. By adjusting the imaging sequence, however, stones can be identified with increased reliability.⁵¹² The sensitivity of MRI (82%) is higher than that of ultrasonography and KUB radiography but less than that of CT.

MANAGEMENT OF STONES

MEDICAL MANAGEMENT OF ACUTE RENAL COLIC

A passing stone is associated with significant pain, disability, and loss of productivity.⁵²² Approximately 50% of patients experiencing acute upper urinary tract stones require surgical intervention.^{523,524} Stone size and location in the urinary tract are the major determinants of the likelihood of spontaneous stone passage, which is higher for distal ureteral calculi compared with proximal and middle locations. Rates are also higher for smaller (<4 mm in diameter) compared with larger stones.^{525,526} Overall, spontaneous passage rates are 12%, 22%, and 45% for proximal, middle, and distal ureteral calculi, respectively, and 55%, 35%, and 8% for stones smaller than 4 mm, 4 to 6 mm, and larger than 6 mm, respectively.^{525,527}

Medical management can facilitate spontaneous passage of ureteral stones. Treatment is necessary until spontaneous passage of the stone, which usually occurs within 48 hours after the onset of acute renal colic.⁵²⁸ During that period, patients will require supportive treatment, comprising pain medications, hydration, and antiemetics.⁵²⁹ In certain circumstances such as signs of urinary tract infection, severe pain not responsive to analgesia, ureteral stones >10 mm, or suspicion of fornix rupture, an immediate urologic intervention may be necessary.

Medical expulsive therapy (MET) has been used as conservative management for the treatment of ureteral stones.^{530,531} Paracetamol (acetaminophen), nonsteroidal anti-inflammatory drugs (NSAIDs), and metamizole serve as primary choices for patients experiencing acute renal colic due to their superior analgesic properties compared with opioids and reduced side effects.⁵³² Nevertheless, each comes with its unique set of potential complications, such as agranulocytosis (with metamizole) and the risk of acute kidney injury (with NSAIDs). For patients with distal ureteral stones seeking conservative treatment options, α -blockers (tamsulosin) or calcium channel blockers (nifedipine) may be beneficial, especially for stones >5 mm.⁵³³ Suspected UA stones, characterized by radiolucency (KUB) or low HU units (CT) and a low UpH, might be treatable through oral chemolysis. To this end, urine is alkalinized using potassium

citrate (preferred) or sodium bicarbonate, aiming for a target UpH between 7.0 and 7.2.⁵³⁴

KIDNEY STONE RECURRENCE PREVENTION

DIETARY MANIPULATION AND LIFESTYLE ADJUSTMENTS

Adherence to dietary modifications and lifestyle adjustments can profoundly influence stone recurrence rates, as shown in a population-based cohort study.⁵³⁵ A diet characterized by high fluid intake (2.5–3.0 L/day); low sodium (<2.3 g/day; 24-hour urinary sodium <100 mmol/day) and animal protein consumption (0.8–1.0 g/kg/day); a balanced ingestion of sufficient dairy products (calcium intake 1000–1200 mg/day); and rich in fruits and vegetables is associated with the lowest risk for incident kidney stones.^{495,536} However, a major limitation in the field has been the scarcity of RCTs to compare the effects of specific dietary manipulations in the prevention of stone recurrence.⁵³⁷

Fluid Intake

Since urine volume is a major determinant of the concentration of lithogenic factors, an increased fluid intake is recommended for all stone formers. The effect of fluid intake was first examined in an RCT that randomized first-time calcium stone formers to two different hydration regimens. Over 5 years, high fluid intake to ensure a urinary volume of approximately 2.5 L/day reduced stone recurrence by 45% compared with the control group that had a urinary volume of approximately 1 L/day.⁴⁹ An observational study in the general population estimated that each 200 mL increase in fluid intake is associated with a 13% reduction in the risk of incident stone events.⁵³⁵ Factors that seem to affect the risk of stone recurrence are not only the amount of fluid consumed but also the type of beverage. In a 3-year RCT, a marginal effect with avoidance of soft drinks was observed.⁵³⁸ Observational evidence suggests that consumption of sugar-sweetened soda, punch, and grapefruit juice is associated with a higher risk of stone formation, whereas consumption of coffee, tea, beer, wine, and orange juice were associated with a lower risk.^{539,540}

Dietary Adjustment

Sodium. Although high dietary sodium intake is primarily associated with hypertension and an elevated risk of cardiovascular diseases, it is also significantly correlated with urinary calcium excretion. As the sodium intake rises, urinary calcium excretion increases proportionately.⁵⁴¹ For every 6 g increment in salt intake, there is an associated 40 mg/day increase in urinary calcium excretion.^{537,542} Several studies have investigated the relationship between dietary salt restriction, calcium excretion, and kidney stone recurrence. The DASH diet, which emphasizes a low-sodium consumption and a high intake of fruits and vegetables, is associated with decreased urinary calcium excretion and a reduced risk of incident kidney stones.^{543,544} These studies underscore the importance of limiting dietary sodium, particularly in patients with hypercalciuria.

Calcium. A high calcium intake is associated with a reduced risk of stone formation in the general population.¹⁸ In an RCT involving men with recurrent calcium oxalate stones

and hypercalciuria, a diet low in calcium (400 mg/day) was compared with a diet high in calcium (1200 mg/day) and low in sodium (<100 mEq/day) and animal protein consumption (50–60 g/day).⁴⁸ After 5 years, the high calcium diet decreased urinary oxalate excretion and resulted in a 50% reduction in stone recurrence. The relationship between dietary calcium intake and urinary oxalate excretion is well documented: Calcium acts as a chelator for oxalate in the intestine. Consequently, a diet low in calcium can elevate the fraction of free oxalate ions in the intestine, promoting its absorption and subsequent urinary excretion.⁵⁴⁵ Unlike calcium intake by natural sources (dairy products and calcium-rich mineral water), calcium supplements not timed with meals increase urinary calcium excretion without beneficial effects on urinary oxalate, thereby increase the risk for stone formation.⁴⁷⁵

Oxalate. The protectiveness of dietary oxalate restriction is debated. Although urinary oxalate excretion correlates with the risk of nephrolithiasis,⁵⁴⁶ dietary oxalate was not found to be a major risk factor for nephrolithiasis in the general population.⁵⁴⁷ The bioavailability of dietary oxalate, and hence urinary oxalate excretion, is strongly influenced by calcium intake. Therefore it is advisable to recommend patients with idiopathic hyperoxaluria and CaOx stones to consume 1000 to 1200 mg of calcium per day, timed with meals, and to avoid any excess of oxalate-rich foods. In patients with enteric hyperoxaluria, dietary oxalate and fat restriction should be imposed in addition to higher total calcium intake from the diet and supplements at mealtime.^{548–550}

Vitamin C intake above the physiologic dose may also increase urinary oxalate excretion by enhancing its hepatic synthesis. Therefore patients with hyperoxaluria and kidney stones should refrain from the use of high-dose vitamin C supplements.^{551,552}

Protein. In a multicomponent dietary RCT, it was shown that a combination of a low-protein (50–60 g/day), low-sodium diet (≤ 100 mEq/day) combined with normal calcium intake (1200 mg/day) significantly lowered the risk of kidney stone recurrence.⁴⁸ This study was performed in adults, and protein restriction in growing children is generally not recommended. The source of protein is also important. In the general population, dairy animal protein is associated with a reduced risk and nondairy animal protein is associated with an increased risk of stones, whereas vegetable protein shows no association with stone risk.⁵⁵³

Fruits and Vegetables. Fruits and vegetables are the primary source of dietary alkali, and 24-hour urinary potassium excretion is an established proxy of daily fruit and vegetable consumption. A reduced intake of fruits and vegetables elevates the risk of stone formation by decreasing urine pH and the urinary excretion of citrate, magnesium, and potassium.^{539,554}

Fructose. Fructose intake has risen massively in the past 50 years.⁵⁵⁵ Fructose stimulates the production of UA, lowers urine pH, and augments urinary excretion of calcium, phosphate, and oxalate.^{556–558} These metabolic changes may increase the risk for formation of both calcium and UA stones. Indeed, in three large observational cohorts of health

professionals, high fructose intake was associated with an increased risk of incident kidney stones in both men and women.⁵⁵⁹ In contrast, nonfructose carbohydrates were not associated with an increased risk.

PHARMACOLOGIC TREATMENT

If dietary manipulation fails to correct metabolic abnormalities and/or patients continue to experience stone recurrence, pharmacologic prevention is recommended. Appropriate treatment should be based on stone phenotype, 24-hour urinary lithogenic risk factors, and patient comorbidities. Pharmacologic treatment is necessary in most cases of recurrent kidney stone formers and in the management of patients with uric acid, cystine, and infection-induced stones.⁵⁶⁰ One major limitation with respect to pharmacologic treatment is the paucity of randomized controlled data.⁵²⁶ There is also no consensus as to whether pharmacologic treatment should be targeted at specific metabolic abnormalities or should be given empirically.^{561–565} The list of commonly used drugs and recommended dosages is summarized in [Table 40.6](#).

PHARMACOLOGIC AGENTS USED

Thiazide Diuretics

Thiazide and thiazide-like diuretics (thiazides) reduce urine calcium excretion and have been the cornerstone of pharmacologic recurrence prevention of kidney stones for more than 50 years.^{495,566,567} Hydrochlorothiazide (HCTZ) is by far the most widely used and best-studied thiazide. In 9 of 11 past trials, thiazides reduced the risk of kidney stone recurrence, with similar efficacy using once- or twice-daily dose regimens.^{565,568–571} However, all these trials had major methodologic limitations.^{564,565,568–576} To address these, a double-blind, randomized, placebo-controlled trial to evaluate incremental doses of HCTZ (12.5-, 25- and 50-mg) in recurrent calcium stone formers with no secondary causes (NOSTONE trial) were recently conducted.⁵⁷⁷ The primary endpoint was a composite outcome of symptomatic or CT-diagnosed radiologic recurrence. Over a period of 3 years, the incidence of recurrence did not differ between groups and there was no relation between hydrochlorothiazide dose and occurrence of a primary endpoint event. Although the secondary endpoint radiologic recurrence (a composite of new stones or stone growth on CT) was numerically less frequent in the 25-mg and 50-mg groups, no difference in the number of new stones formed among groups was observed. In addition, recurrence rates were similar in HCTZ-treated and placebo-treated patients who were stone free at baseline. Adverse events such as hypokalemia, gout, new-onset diabetes mellitus, skin allergy, and increasing plasma creatinine were more common among patients who received HCTZ. Together, these results indicate that HCTZ is not efficiently preventing recurrence of kidney stones yet is associated with significant adverse events.

In line with lack of efficacy of HCTZ on clinical outcomes, no consistent changes in urine calcium oxalate relative supersaturation for CaOx and CaP were observed between groups, although the reduction in urinary calcium (9%–17% compared with baseline) was lower than expected (20%–40%) based on previous trials.^{571,578,579} Apart from

Table 40.6 Commonly Used Drugs in the Treatment of Nephrolithiasis

Class of Drugs	Molecule	Indication	Dosage
Calcium-Containing Stones			
Thiazide diuretics	Hydrochlorothiazide	Hypercalciuria	50 mg/day
	Indapamide	Hypercalciuria	2.5 mg/day
	Chlorthalidone	Hypercalciuria	25 mg/day
Alkali supplements	Potassium citrate	Hypocitraturia	20-60 mEq/day, divided into 2-4 times
		Acidic UpH	
Xanthine oxidase inhibitors	Allopurinol (first choice)	Hyperuricosuria	100-300 mg/day
	Febuxostat		
Calcium supplements	Calcium carbonate	Hyperuricosuria	40--80 mg/day
		Secondary hyperoxaluria	500-1500 mg/day, divided into 2-3 times during meals
Uric Acid Stones			
Alkali supplements	Potassium citrate	To maintain UpH between 6.0 and 7.0	20-80 mEq/day, divided into 2-4 times
Xanthine oxidase inhibitors	Allopurinol (1st choice)	Hyperuricosuria	100-300 mg/day
	Febuxostat	Hyperuricosuria	40--80 mg/day
Cystine Stones			
Alkali supplements	Potassium citrate	To maintain UpH >7.0	20-80 mEq/day
Thiol compounds	Tiopronin	Urinary cystine >1000 mg/day	750-1500 mg/day
Struvite Stones			
Antibiotic therapy	Nitrofurantoin monohydrate/trimethoprim-sulfamethoxazole	Preoperatively/perioperatively, to reduce the risk urosepsis	Dosages must be adapted for each patient and involved bacteria
Urease inhibitors	Acetohydroxamic acid	In adjunct to antibiotics, in patients with remaining stone fragments and no other surgical options	500 mg/day, divided into 2 doses
Acidifying agents	L-methionine	Inappropriately high urine pH (>7.0)	1500-3000 mg/day

overestimation of treatment effects of HCTZ on calciuria in past studies due to methodologic deficiencies, the relatively high sodium consumption likely contributed to this observation. In the NOSTONE trial, sodium intake was higher than recommended (4.2–4.6 g/day) despite repeated dietary instructions. This observation is consistent with results of past thiazide trials reporting similar sodium intake during follow-up,^{571,572} demonstrating the difficulty to achieve a sustained reduction of sodium intake. Additionally, urinary citrate tended to be lower in patients receiving HCTZ. Reduction of urine citrate is thought to be a consequence of thiazide-induced hypokalemia, resulting in intracellular acidosis, thereby stimulating citrate reabsorption in the proximal tubule⁵²⁶ and thus partially counteracting the beneficial effects of these molecules on urinary calcium.

Thiazide-like diuretics, such as indapamide and chlorthalidone, are more potent and have a significantly longer half-life compared with hydrochlorothiazide.^{564,572} Consequently, one might presume they offer better stone prevention. However, this assertion lacks robust RCT evidence, and no head-to-head comparison of different thiazides for kidney stone recurrence prevention or for the

established proxies of recurrence risk, urine RSR CaOx, and CaP has ever been performed.

Hence we suggest avoiding HCTZ for the sole indication of stone prevention. In patients with active calcium stone disease, unresponsiveness to dietary manipulation and citrate supplementation, and no further treatment options, long-acting thiazides may still be used.

Alkali Treatment

Alkali may be used to prevent recurrence of UA or calcium-containing kidney stones. No RCT has ever been performed to assess the efficacy of alkali treatment in UA stone formers, but anecdotal evidence indicates effective suppression of UA stone formation if a urine pH >6 can be achieved. For the prevention of calcium-containing kidney stones, several small RCTs have been conducted thus far with alkali—mostly administered as potassium citrate.^{561,580-585} A meta-analysis of these trials concluded that alkali therapy decreases the incidence of new stone formation (relative risk 0.26, 95% CI 0.10–0.68), compared with control treatment or placebo.⁵⁸⁶ In addition to reducing the incidence of new stones, alkali treatment lowered stone growth and increased urinary citrate excretion. However, adverse events with alkali treatment

were found to be common, especially gastrointestinal side effects. As a result, the dropout rate due to adverse events was significantly higher in patients receiving alkali compared with patients receiving placebo or usual care. The systematic Cochrane review further concluded that the quality of the reported literature remains moderate to poor and that there is a need for a well-designed statistically powered multicenter RCT to answer relevant questions concerning the efficacy of alkali in the prevention of calcium-containing kidney stones. Another potential side effect of alkali treatment is the risk of CaP stone formation due to an increase in urine pH. Yet observational evidence indicates that alkali therapy may prevent stone recurrence, even in patients with alkaline urine pH due to distal RTA.^{169,587,588}

Although sodium bicarbonate may offer the same degree of urinary alkalization when used in an equivalent dose to potassium alkali, it may increase the risk of calcium stone formation because of a sodium-induced increase in urinary calcium and promotion of monosodium urate-induced CaOx crystallization.^{263,589} The initial recommended dose for alkali is 30 to 40 mEq/day, typically divided equally two to four times daily. Because unduly acidic urine is the predominant feature of patients with UA nephrolithiasis, alkali therapy with potassium citrate is first-line treatment in these patients and higher doses may be needed to maintain UpH between 6.0 and 7.0.²⁷⁷ Sodium alkali may be used in patients with renal insufficiency to avoid hyperkalemia or in patients with volume depletion due to gastrointestinal fluid losses (e.g., in patients with ileostomy).

Xanthine Oxidase Inhibitors

Allopurinol is used for the treatment of hyperuricosuric calcium stone formers rather than UA stone formers. One RCT in hyperuricosuric calcium stone formers, comparing allopurinol versus placebo,⁵⁹⁰ demonstrated significantly decreased stone events with allopurinol treatment. The efficacy of allopurinol monotherapy in the treatment of hyperuricosuric CaOx stone-forming patients with multiple metabolic abnormalities is less evident.⁵⁹¹ The effective dose is 300 mg/day, and adjustments are needed in patients with impaired kidney function because allopurinol is primarily excreted by the kidney.

Febuxostat, a nonpurine xanthine oxidase inhibitor analog metabolized by the liver,^{592,593} can be considered as an alternative to allopurinol. In a 6-month, double-blinded RCT, hyperuricosuric calcium stone formers were treated with 80 mg/day of febuxostat, 300 mg/day of allopurinol, or placebo. Febuxostat reduced 24-hour urinary UA significantly more than allopurinol, but there was no change in stone size or number over this period of time.⁵⁹⁴ Therefore due to insufficient follow-up time, no definitive conclusion can be drawn.

Cystine Chelation Therapy

Thiol derivatives act as chelating agents and are therefore commonly used in patients unresponsive to conservative management with severe cystinuria (>1000 mg/day). These drugs reduce dimeric cystine into two cysteine monomers to form a highly soluble compound with the drug.⁵⁹⁵ Two main thiol compounds are available, a disulfide compound called D-penicillamine and α -mercaptopyrionylglycine (tiopronin). In two studies, pharmacologic treatment with

D-penicillamine or tiopronin was found to be superior to hydration and alkalization alone in reducing stone events.^{596,597} However, to date, no RCTs have demonstrated the superiority of pharmacologic treatment over placebo in patients with cystinuria.

Tiopronin has a lower incidence of side effects compared with D-penicillamine, and it is therefore used more often.⁵⁹⁸ Side effects include GI irritation, abnormalities in hepatic enzymes, pancytopenias, allergic skin reactions, and proteinuria.⁵⁹⁸ Effective therapy should be aimed at maintaining a urinary cystine concentration at 250 mg/L, although there is variability among individual patients. However, to date, there has been no clinical study exploring whether such measurement will avert the complications of cystine stone formation.^{354,599,600} One important caveat is that many commercial assays for urinary cystine measurement do not differentiate between a drug-cysteine complex and unbound cystine.

Pharmacotherapy of Infection-Related Stones

According to guidelines, surgical treatment is the best management for infectious stones. Struvite staghorn calculi generally require surgical removal and, if not properly treated, may require nephrectomy.^{601,602} However, medical treatment can be used in those with significant comorbidities who are not amenable to surgery, to preserve renal functional reserve, and also to prevent stone recurrence after successful stone removal. These treatments include antibiotics, dissolution therapy, urease inhibitors, urinary acidification, dietary modification, and other supportive measures. Antibiotics are valuable during both preoperative and perioperative periods to prevent urinary sepsis from surgical procedures. Urease-producing microorganisms, by converting urea to ammonia, play a key role in supersaturation of urine with respect to ammonium magnesium phosphate (struvite), which usually coexists with calcium carbonate apatite stones, because both stone types are favored in an alkaline environment.⁶⁰³ When combined with antimicrobial therapy, urease inhibition provides palliation for patients who cannot undergo definitive surgical management, although it cannot eradicate existing stones.^{604,605} The urease inhibitor acetohydroxamic acid is the only U.S. Food and Drug Administration–approved urease inhibitor that can be administered orally to inhibit urease enzyme activity.^{606,607} However, acetohydroxamic acid can confer serious GI, neurologic, hematologic, and dermatologic side effects in up to 20% of patients, which tends to resolve after discontinuation.⁶⁰⁶⁻⁶⁰⁸ Because urine alkalinity (pH >7.2) promotes the formation of struvite and carbonate apatite stones, urinary acidification has been also used. L-Methionine, which imposes an acid load (is a sulfur-containing amino acid metabolized to H₂SO₄), can lower urine pH. Since decreasing urine pH from 6.5 to 5.7 increased the dissolution rate of struvite stones in vitro,⁶⁰⁹ it was suggested that oral intake of L-methionine (1500 to 3000 mg/day) might be used for the dissolution of infectious stones, although there has been no trial to prove this yet.⁶⁰⁹

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